



Cite this: *Soft Matter*, 2015,
11, 8550

Polyelectrolyte brushes: theory, modelling, synthesis and applications

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Polyelectrolyte (PE) brushes are a special class of polymer brushes (PBs) containing charges. Polymer chains attain "brush"-like configuration when they are grafted or get localized at an interface (solid-fluid or liquid-fluid) with sufficiently close proximity between two-adjacent grafted polymer chains – such a proximity triggers a particular nature of interaction between the adjacent polymer molecules forcing them to stretch orthogonally to the grafting interface, instead of random-coil arrangement. In this review, we discuss the theory, synthesis, and applications of PE brushes. The theoretical discussion starts with the standard scaling concepts for polymer and PE brushes; following that, we shed light on the state of the art in continuum modelling approaches for polymer and PE brushes directed towards analysis beyond the scaling calculations. A special emphasis is laid in pinpointing the cases for which the PE electrostatic effects can be de-coupled from the PE entropic and excluded volume effects; such de-coupling is necessary to appropriately probe the complicated electrostatic effects arising from pH-dependent charging of the PE brushes and the use of these effects for driving liquid and ion transport at the interfaces covered with PE brushes. We also discuss the atomistic simulation approaches for polymer and PE brushes. Next we provide a detailed review of the existing approaches for the synthesis of polymer and PE brushes on interfaces, nanoparticles, and nanochannels, including mixed brushes and patterned brushes. Finally, we discuss some of the possible applications and future developments of polymer and PE brushes grafted on a variety of interfaces.

Received 6th August 2015,
Accepted 7th September 2015

DOI: 10.1039/c5sm01962a

www.rsc.org/softmatter

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1 Introduction

Polymer brushes (PBs)^{1–10} are assemblies of polymer chains tethered or grafted by one of their extremities to a surface



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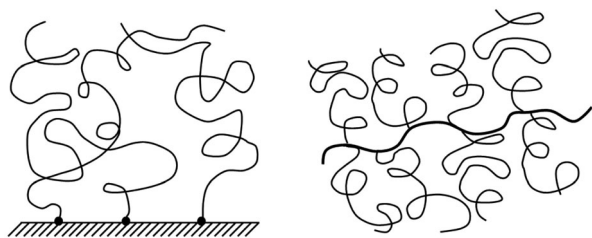


Fig. 1 Schematic of polymer brushes grafted at the solid–liquid interface (left) and the backbone of another polymer (right).

(Fig. 1a) or to the backbone of another polymer chain, which is referred to as the bottle brush polymer (Fig. 1b). PBs are a special class of polymers distinguished by their tethered or grafted state at interfaces. Typically this grafting is sufficiently dense so as to ensure that the polymer chains, instead of attaining the classical random-coil arrangement, are forced to stretch away from the adhering surface like a brush. Therefore, most intuitively, grafting density is one of the central parameters in the context of PBs.



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Depending on the grafting density and the nature of interaction between the polymer chains, three distinct regimes are possible (see Fig. 2). In both Fig. 2a and b, the grafting density is below a critical value (this value is discussed later); this ensures that a specific grafted chain is not influenced by its neighbouring grafted chains, and therefore this chain does not attain a brush-type configuration to regulate its interactions with the adjacent chains. Fig. 2a shows a situation where the polymer chain has a strong affinity to get adsorbed on the substrate surface, resulting in a flat “pancake-like” conformation. On the other hand, in a sparsely grafted system if the polymer chain does not adsorb to the surface a “mushroom”-like conformation results (see Fig. 2b). Fig. 2c represents a situation where the grafting density is greater than the critical value, and the chains, in order to avoid overlapping, stretch in a direction normal to the substrate surface, resulting in a brush. This stretching can ensure, depending on the chain size (*i.e.*, number of monomers in the polymer chain) N , that the length of a grafted polymer in “brush” confirmation is substantially larger than the radius of gyration of the unstretched chain. Such a situation results in unique and often exotic properties such as multifunctional and stimuli responsive surfaces, based on either the presence of more than one type of brush molecules or based on the ability of the brush molecules to change their conformation and thus regulate or adjust their interaction with their surroundings. In the simplest case, the interaction between a brush and the surrounding might simply alter between attractive (switch on) and repulsive (switch off), depending on various simple stimuli such as the type of solvent, pH of the surrounding medium, magnitude of the electrostatic forces and so on. Such a simple effect can have a significant influence on properties like the coefficient of friction, work of adhesion, or wetting, which in turn can modulate cell adhesion and growth, protein adsorption, flow incitation, wetting and dewetting, drug release, control of membrane permeability *etc.* of a surface. Such a modulation in surface properties with grafted brush molecules is completely distinct from the typical behaviour of flexible polymer chains in solution, where the chains exhibit random-walk configuration.

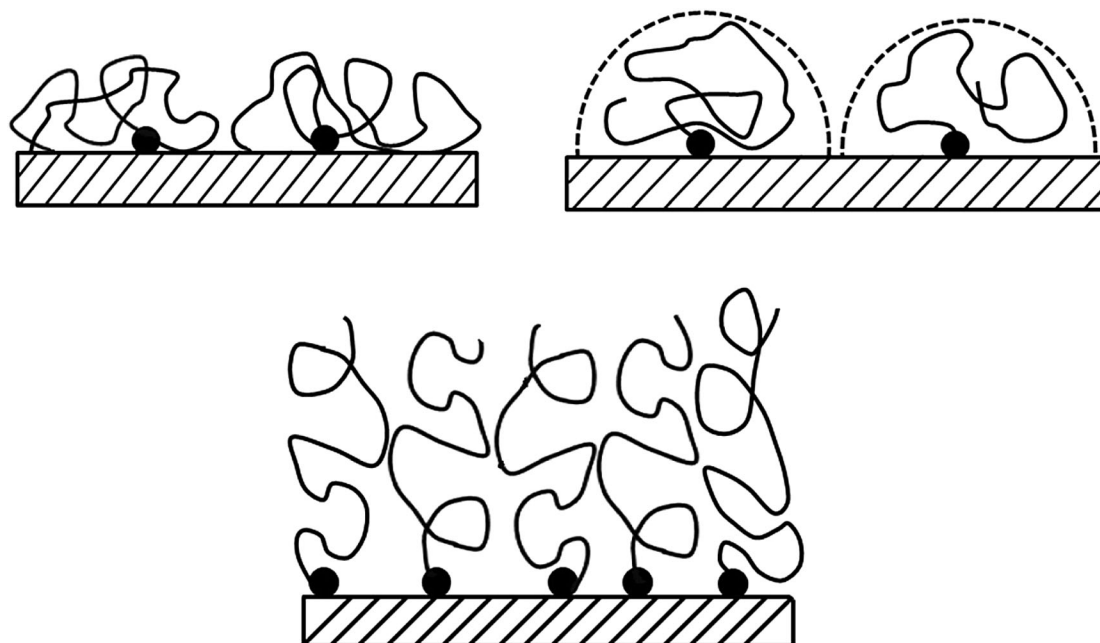


Fig. 2 Schematic of the grafted polymer layers in “pan-cake” configuration (left, top row), “mushroom” configuration (right, top row), and “brush” configuration (bottom row).

Most importantly, these stretched chains exist under equilibrium conditions without any physical confinement or external field being applied, though at the molecular level, PBs are relatively less organized because of their tendency to stretch to their maximum.¹¹ The process of stretching results in a large loss of conformational entropy, and is opposed by the elastic free energy of the chains. Several excellent reviews are already present covering various aspects of polymer brushes and their applications.^{1–10}

In this review we will focus particularly on polyelectrolyte (PE) brushes, which refer to long PE chains grafted densely to a surface (see Fig. 3a). PE brushes, therefore, are PBs containing charges. The presence of charges on the PBs makes them far more responsive than the uncharged PBs. The key feature of any PE brush is a strong confinement of the counter-ions within the brush layer that leads to strong inter segment repulsions and the

electrostatic interactions which are absent in uncharged brushes. While the theoretical development of the science of PE and polymer brushes began several decades ago with the pioneering contributions of Alexander, de Gennes, and Milner,^{12–17} the ability to create nano-functionalized interfaces (such as nanochannels or nanoparticles) by programming the interfaces with polymer or PE grafting (see Fig. 3b and c) and making use of these interfaces in a large number of applications has evoked renewed interest in polymer or PE-grafted surfaces. These applications range from PE-grafted-nanoparticle-driven targeted drug delivery, oil recovery, nanocomposite fabrication, water harvesting as well as the fabrication of ionic diodes and rectifiers, flow-control valves, and sensors for ions and bio-molecules using pH-responsive PE-grafted nano-fluidic channels.

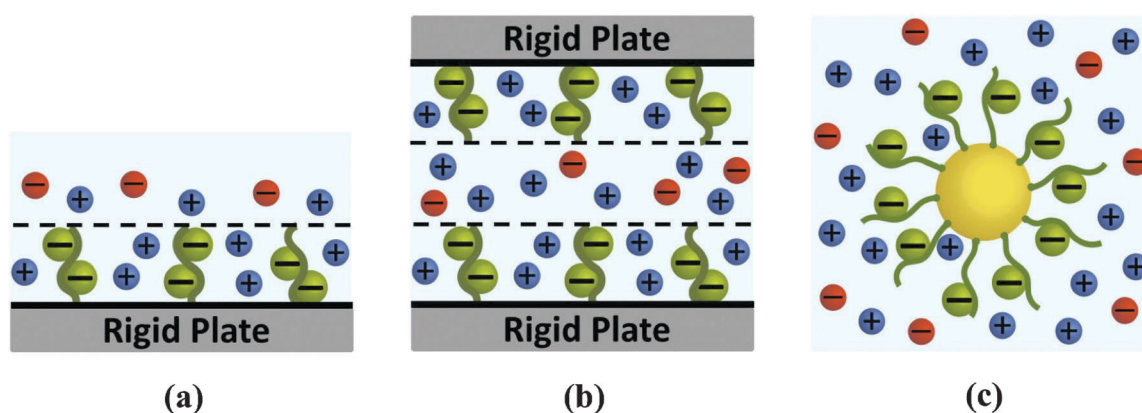


Fig. 3 Schematic showing (a) interfaces, (b) nanochannels, and (c) nanoparticles grafted with negatively charged PE brushes. The charges on the PE brushes are screened by the electrolyte ions forming the Electric Double Layer or EDL.

Fundamentals of polymer-grafted systems as well as the large number of applications of these systems have been well documented in several other previous review articles.^{1–10} In this context, the present review sheds light on the following key areas that have remained relatively uncharted in these reviews: (a) description of PE electrostatic effects for reactive PEs for cases where the electrostatic effects can be decoupled from the elastic and the excluded volume effects of the PE molecules; (b) overview of the methodology to capture the PE electrostatic effects when such decoupling is not possible; (c) electrokinetic transport at the PE-grafted interfaces; (d) synthesis and characterization of a PE grafted system; (e) applications and future developments of PE grafted surfaces.

The present review is organized as follows: in Section 2, we provide a brief review of the basic fundamentals of the modelling of polymer brushes. In Section 3, we analyze the electrostatics of the PE brushes for cases when the electrostatics may or may not be decoupled from the elastic and the excluded volume effects. In Section 4, we describe the transport of liquids and ions at these PE-grafted interfaces. In Section 5, we review the mesoscopic and the atomistic simulation approaches employed to study the behaviour of these polymer and PE brushes. In Section 6, we review the historical development and the state of the art in fabricating interfaces with polymer and PE brushes. In Section 7, we review the different applications of these brushes, ranging from the fabrication of superhydrophobic surfaces and controlling wetting dynamics to using PE-grafted nano-interfaces in a myriad of disciplines. In Section 8, we discuss some of the future applications of PE brushes. Finally, in Section 9, we end with concluding remarks.

2 Mathematical description of polymer brushes

2.1 Scaling laws for polymer brushes

Grafted polymer chains attain brush-like configuration when their grafting density σ (*i.e.*, per unit area there are σ chains grafted or affixed to the solid surface) is above a critical value (σ^*). This critical value can be obtained as follows. The separation distance between two adjacent grafted polymer chains, quantified as $\sigma^{-1/2}$, must be such that these two adjacent grafted chains start interacting with each other. This is possible only when this distance of separation is smaller than the chain radius $R \sim aN^\nu$ (where a is the Kuhn length of the polymer). Therefore the critical grafting density σ^* should be such that the corresponding distance of separation is equal to this chain radius. Hence:

$$\sigma^{*-1/2} \sim aN^\nu \Rightarrow \sigma^* \sim a^{-2}N^{-2\nu}. \quad (1)$$

For an ideal polymer chain (*i.e.*, where there are no interactions between monomers), $\nu = 1/2$; on the other hand, when the chain is in a good solvent such that the monomer–solvent interactions are more favourable than the monomer–monomer interactions, $\nu = 3/5$. We shall primarily work with the latter

case. Accordingly, from eqn (1) we can estimate the critical grafting density (σ^*) as:

$$\sigma^* \sim a^{-2}N^{-6/5}. \quad (2)$$

While eqn (2) provides the mathematical condition for ensuring the formation of PBs in a good solvent, the involved physics has been beautifully explained by Milner.¹⁷ Here we briefly summarize that explanation. In a good solvent, the grafted polymer chains demonstrate two mutually contesting tendencies, namely adoption of random-walk configurations and getting maximum exposure and wetting by the solvent. While the first tendency maximizes the configurational entropy and favours short-length, dense brushes, taller and sparse brushes better satisfy the second tendency. For grafted polymer systems, in case the distance between two adjacent grafted molecules ($\sigma^{-1/2}$) is much smaller than $\sigma^{*-1/2}$, these above two tendencies cannot be satisfied simultaneously. Consequently, the chain stretches out as a “brush” (or in other words, chooses a height d) so as to minimize the total energy by balancing these two tendencies. Mathematically, this can be expressed by writing the free energy change per grafted polymer molecule (ΔF_{PB}) as:¹⁷

$$\Delta F_{PB} = \Delta F_{PB,ent} + \Delta F_{PB,VE} \sim k_B T \left[\frac{3d^2}{2Na^2} + wN \left(\frac{N\sigma}{d} \right) \right]. \quad (3)$$

In eqn (3), $k_B T$ is thermal energy, w is the excluded volume parameter, $\Delta F_{PB,ent} \sim k_B T \frac{3d^2}{2Na^2}$ is the entropic contribution originating from chain stretching under the assumption that the polymer chains are springs with spring constant $k_B T/Na^2$, and $\Delta F_{PB,VE} \sim k_B T wN \left(\frac{N\sigma}{d} \right)$ is the excluded volume contribution to the free energy change caused by the reduction of the contact of the chain with the solvent molecules. The equilibrium height $d_{eq,PB}$ can be obtained by minimizing eqn (3) with respect to d , yielding:

$$d_{eq,PB} \sim N(2w\sigma a^2/3)^{1/3}. \quad (4)$$

Eqn (4) shows that the thickness of the polymer brushes scales linearly with the polymer size N ; this manifests the distinctive “stretching” behaviour of the brushes, since for an unstretched chain, the chain dimension scales as N^ν ($\nu = 3/5$ for a good solvent).

2.2 Monomer distribution profiles of polymer brushes

The scaling laws described in the previous subsection cannot provide a quantification of the monomer density distribution along the depth of the grafted polymer molecule in the brush-like configuration. This distribution has been obtained by a detailed theoretical model provided separately by Milner *et al.*,^{17–19} Skvortsov *et al.*,²⁰ Zhulina *et al.*²¹ Here we provide an illustrative description of this model.

Just like the derivation of the scaling laws, here too, the starting point is expressing the total free energy of a single grafted polymer molecule as a sum of the elastic and the excluded volume contributions. The main difference, however, is that for the present case, these energy contributions are expressed as a function of the dimensionless monomer distribution $[\varphi(y)]$.

Therefore, considering the polymer chain to be flexible, we may write:

$$\begin{aligned}\Delta F_{\text{PB}} &= \Delta F_{\text{PB,ent}} + \Delta F_{\text{PB,VE}} \\ &= \frac{3k_{\text{B}}T}{2a^2} \int_0^d g(y') dy' \int_0^{y'} E(y, y') dy + \frac{k_{\text{B}}T}{\sigma a^3} \int_0^d f_{\text{conc}}[\varphi(y)] dy.\end{aligned}\quad (5)$$

In eqn (5), $g(y')$ is the normalized density of the free end of a grafted polymer chain, $E(y, y')$ is a function that denotes the extent of the stretching of a polymer chain at a height y , with the free end of the chain being located at a height y' (here $y' > y$ always), and f_{conc} is the dimensionless free energy density caused by the excluded volume interactions.

This energy needs to be minimized in presence of the conditions:

$$N = \int_0^{y'} \frac{dy}{E(y, y')}, \quad (6)$$

$$N = \frac{1}{\sigma a^3} \int_0^d \varphi(y) dy. \quad (7)$$

Also $\varphi(y)$ is related to the functions g and E as:

$$\varphi(y) = a^3 \sigma \int_y^d \frac{g(y') dy'}{E(y, y')}. \quad (8)$$

Accounting for the constraints [eqn (6) and (7)] the free energy can be expressed in terms of the Lagrange multipliers (λ_3 and λ_4) as:

$$\begin{aligned}\Delta F_{\text{PB}} &= \frac{3k_{\text{B}}T}{2a^2} \int_0^d g(y') dy' \int_0^{y'} E(y, y') dy + \frac{k_{\text{B}}T}{\sigma a^3} \int_0^d f_{\text{conc}}[\varphi(y)] dy \\ &\quad + \lambda_3 \left[\frac{1}{\sigma a^3} \int_0^d \varphi(y) dy - N \right] + \int_0^d \lambda_4(y') dy' \left[\int_0^{y'} \frac{dy}{E(y, y')} - N \right].\end{aligned}\quad (9)$$

This equation is subsequently minimized using variational approach, leading to (see Zhulina and Borisov²² for detailed derivation) the following two equations:

$$\frac{3g(y')}{2a^2} - \frac{\lambda_4(y')}{E^2(y, y')} - \left(\frac{\delta f_{\text{conc}}}{\delta \varphi} + \lambda_3 \right) \frac{g(y')}{E^2(y, y')} = 0, \quad (10)$$

$$\int_0^{y'} dy \left[\frac{3E(y, y')}{2a^2} + \left(\frac{\delta f_{\text{conc}}}{\delta \varphi} + \lambda_3 \right) \frac{1}{E(y, y')} \right] = 0. \quad (11)$$

From eqn (10), we can write:

$$E^2(y, y') = \frac{2a^2}{3} \left[\frac{\lambda_4(y')}{g(y')} + \left(\lambda_3 + \frac{\delta f_{\text{conc}}}{\delta \varphi} \right) \right] = U_1(y') - U_2(y), \quad (12)$$

where

$$U_1(y') = \frac{2a^2}{3} \frac{\lambda_4(y')}{g(y')}, \quad (13)$$

and

$$U_2(y) = \frac{2a^2}{3} \left(-\lambda_3 - \frac{\delta f_{\text{conc}}}{\delta \varphi} \right). \quad (14)$$

Since $E(y, y) = 0$,^{21,22} we can write:

$$U_1(y) = U_2(y) = U(y). \quad (15)$$

Consequently, we can write:

$$E(y, y') = \sqrt{U(y') - U(y)}. \quad (16)$$

The normalization condition given by eqn (6) serves as an integral equation for $U(y')$. One can check that this integral equation is satisfied if

$$U(y) = \frac{\pi^2 y^2}{4N^2}. \quad (17)$$

Consequently,

$$E(y, y') = \frac{\pi}{2N} \cdot \sqrt{y'^2 - y^2}. \quad (18)$$

Now we can re-write eqn (11) using eqn (14) and (15) as:

$$\int_0^{y'} dy \left[E(y, y') - \frac{U(y)}{E(y, y')} \right] = 0. \quad (19)$$

Eqn (19) is equally satisfied with these stated forms of $U(y)$ and $E(y, y')$. To obtain $\varphi(y)$ we can employ eqn (14). But prior to that we would need the functional dependence of f_{conc} on $\varphi(y)$. Considering virial expansion for the density of free energy of volume interactions, we can write:

$$f_{\text{conc}}[\varphi(y)] \approx \nu_2 \varphi^2 + \nu_3 \varphi^3 + \dots \quad (20)$$

Considering only the first two terms of the expansion of $f_{\text{conc}}[\varphi(y)]$, we can use eqn (15), (17) and (20) in eqn (14) to obtain a quadratic equation in $\varphi(y)$, which can be solved to express $\varphi(y)$ as:

$$\varphi(y) = \frac{\nu_1}{3\nu_2} \left[\sqrt{1 + \kappa^2(\lambda_5 - y^2)} - 1 \right], \quad (21)$$

where

$$\kappa^2 = \frac{9\pi^2 \nu_2}{8N^2 a^2 \nu_1^2}, \quad (22)$$

$$\lambda_5 = -\lambda_3 - \frac{8a^2 N^2}{3\pi^2}. \quad (23)$$

Eqn (21) is the central result of this derivation, providing a quantification of the dimensionless monomer distribution $\varphi(y)$. The value of the indeterminate multiplier λ_5 appearing in eqn (21) can be obtained by using the expression for $\varphi(y)$ [see eqn (21)] in the normalization condition provided by eqn (7). This results in a transcendental equation of λ_5 in terms of the brush height d , i.e.,

$$\begin{aligned}\frac{N\sigma a^3}{d} &= \frac{\nu_1}{3\nu_2} \left[\frac{1}{2} \sqrt{1 + \kappa^2(\lambda_5 - d^2)} - 1 \right] \\ &\quad + \frac{1 + \kappa^2 \lambda_5}{2\kappa^2} \frac{\pi}{\sqrt{8\nu_2 a N d}} \sin^{-1} \left[\sqrt{\frac{\kappa^2 d^2}{1 + \kappa^2 \lambda_5}} \right]\end{aligned}\quad (24)$$

Once these expressions of $E(y, y')$ and $\phi(y)$ are known, one can use eqn (8) as the integral equation to obtain $g(y')$ as:²¹

$$g(y') = \frac{y'\nu_1}{3\sigma\nu_2Na^3} \left[\frac{(1 + \kappa^2(\lambda_5 - d^2))^{1/2} \operatorname{sgn}(\nu) - 1}{(H^2 - y'^2)^{1/2}} \right] + \frac{y'\nu_1}{3\sigma\nu_2Na^3} \left[\frac{\kappa}{2} \left(\frac{\pi}{2} + \sin^{-1} \left[\frac{2d^2 - y'^2 - \kappa^{-2} - \lambda_5}{\lambda_5 + \kappa^{-2} - y'^2} \right] \right) \right]. \quad (25)$$

Given the final expressions for $E(y, y')$, $\phi(y)$, $g(y')$, and $f_{\text{cond}}[\phi(y)]$, we can obtain the free energy from eqn (5). This free energy will now have two variables, namely d and λ_5 . At equilibrium $d = d_{\text{eq}}$ and $\lambda_5 = \lambda_0$. One equation governing d_{eq} and λ_0 can be obtained by replacing d by d_{eq} and λ_5 by λ_0 in eqn (24). The other equation will come from the condition:

$$\left(\frac{\partial F}{\partial d} \right)_{d=d_{\text{eq}}, \lambda_5=\lambda_0} = 0, \quad (26)$$

where F is provided by eqn (5).

2.3 Polymer brushes for other scenarios

In this subsection, we discuss the behaviour of the polymer brushes for situations that are different from that assumed in Sections 2.1 and 2.2. Such differences can arise due to the change in the quality of the solvent (which in turn dictates the relative difference in the monomer–monomer and monomer–solvent interactions), nature of the substrate, and other effects (e.g., chain stiffness, lack of homogeneity of PBs, *etc.*).

Effects of variation in solvent quality. Solvent quality dictates the nature of interactions between the solvent and the monomer of the polymer chains. Here we shall only briefly discuss certain key issues related to PBs in a “poor” solvent. A “poor” solvent, with respect to a given polymer, is defined as a solvent in which the different parts of the polymer encounter a solvophobic interaction enforcing them to drive away the solvent in their vicinity and attract each other.^{23,24} The result is a collapse of the polymer chain (or an equivalent coil-to-globule transition of the polymer chain).^{23–25} The corresponding radius of the polymer chain is $R \sim N^{1/3}$, as compared to $R \sim N^{3/5}$ for the polymer in a good solvent. Typically the crossover state between the “good” and the “poor” solvent behaviour is called the θ state, characterized by the theta temperature T_θ . The polymer collapse is associated with a phase transition, governed by the corresponding critical temperature T_c for the given polymer–solvent system.

The most remarkable aspect of PBs in a poor solvent is that very much like the PBs in a good solvent, $d \sim N$ (d is the brush height).^{21,26,27} This is specially interesting given the large difference in scaling of the radius R of the polymers (in solution) with N between the cases of “good” and “poor” solvents. The calculations invoked for the scaling analysis of the polymer brushes in a poor solvent are very much similar to those illustrated in Section 2.1. Here too one considers the balance of the elastic and the excluded volume energies; the key thing is that the excluded volume energy is expressed as in eqn (21) and

one considers the second virial coefficient as negative for a “poor” solvent. Zhulina *et al.*²⁷ extended this calculation beyond the scaling analysis and probed the internal structure of the PB layer (e.g., monomer and degree of extension distribution within the layer; similar to that illustrated in Section 2.2) in a “poor” solvent. All these calculations also established that the collapse transition of the PBs in a “poor” solvent cannot be quantified as a true phase transition.^{26,27} There are also several insightful studies that describe the equilibrium configuration of PBs in a mixture of “good” and “poor” solvents by invoking the idea of partial solvent remixing.^{28,29}

In this context, it is also worthwhile to mention the studies on the configuration of the PBs in a solvent that is itself a polymer solution.^{14,30} The key parameters concerning these studies are the relative length and nature of the polymers in the “brush” state and in the bulk. Analyses of these problems are provided in terms of the identification of regimes dictated by the extent to which the polymer solvent gets expelled from the vicinity of the brushes and the extent to which the polymer solvent can manoeuvre the interaction forces between two layers of PBs separated by this polymer solvent.³¹

Effects of variation in geometry and nature of the substrate.

While the major amount of theoretical studies concerning PBs consider planar grafting (or equivalently, planar substrates), there has been ample research involving non-planar geometries with finite curvature effects. Daoud and Cotton³² were among the first groups to illustrate the curvature effects on PBs when they studied tethered star polymer systems. PBs grafted to spherical geometries, often denoted as spherical brushes, exhibit a universal density profile that can be demarcated into three separate zones. The first zone or the innermost zone shows a densely packed core; the second zone or the intermediate zone is characterized by the absence of correlation effects and in this zone the monomer density exhibits a r^{-1} variation; finally, in the third zone or the outermost zone, the correlation effects are significant with the monomer density showing a $r^{-4/3}$ variation. Other related studies highlighting the behaviour of spherical brushes or brushes with a finite curvature in “good” as well as “poor” solvents have been performed using mean-field theories and molecular simulations.^{27,33–35}

One of the key motivations for understanding the behaviour of non-planar brushes has been unravelling different applications in biomedical systems. Perhaps the biggest related example is the drug delivery based on nanoparticles grafted with PBs. Such engineering of nanoparticles by the use of PBs ensures a significantly more enhanced efficiency of the targeted drug delivery.^{36–38} A related issue – also of fundamental importance in analyzing the role of substrates in PBs – is the behaviour of these brushes when they are grafted on liposome surfaces for targeted drug and gene deliveries.^{39,40} Such a design will imply that the polymer brushes are now grafted on penetrable lipid bilayers (since the liposomes are made up of lipid-bilayer membranes). This system, therefore, should be treated as a multilayer formed by the lipids and the polymers. The most important aspect of this bilayer is that the distribution of polymers and its nature (adsorbing or non-adsorbing) significantly influence the lipid membrane curvature.^{41–45}

There are several studies that investigate the behaviour of polymer brushes grafted not at any solid interface, but at a liquid–fluid or liquid–liquid interface.^{2,8,46–48} Very often these brushes are di-block co-polymer brushes, where the different blocks exhibit different affinities for the individual solvents. Such localization is important for controlling the properties of the liquid–fluid and liquid–liquid interfaces for a host of different applications.

Influence of other effects. In addition to the influence of the solvent and substrate qualities, issues such as the chain stiffness,^{49–55} homogeneity of the PBs (*i.e.*, whether the PBs are homopolymer or copolymer brushes),^{4,56} *etc.* have also received considerable attention. Here we refrain from discussing much details about these issues; interested readers can follow several excellent and comprehensive studies on these topics identified here.

3 Mathematical description of PE brushes

PEs are polymers containing charges. The charges can be present along the entire backbone of the PEs (*e.g.*, poly-acrylic acid or PAA), or they may be present only at specified locations on the PE molecule (*e.g.*, tip of the PE molecule). The presence of such charges not only leads to new addition to the total free energy of the PE brush, but also affects the surrounding solvent in a manner that is not possible for the polymer brushes. For example, the PE charges trigger a double layer of charge (known as the Electric Double Layer or EDL) surrounding the charged PE brush. Most interestingly, this EDL extends both inside and outside the PE brush (see Fig. 3). Therefore, it becomes inappropriate to analyze only the free energy change of the PE brush; rather the analysis must involve, in addition to the energy change of the PE brush, the energy change associated with this EDL formation. This makes the PE brushes much more non-trivial than the PBs.

3.1 Scaling laws for PE brushes: conditions for decoupling of various effects

Here we provide a scaling description of the PE brushes, with a special emphasis on the alterations in case the PE brushes are grafted on the inner walls of a nanochannel. This derivation and those that are provided in the following two subsections are adapted from our recent study.⁵⁷ For a single PE brush, the relevant free energy change is the sum of the free energy changes associated with the PE molecule and the surrounding EDL. Of course, we assume that the PE brushes are present in a solvent that allows sustaining of the EDL. Therefore, we may write the net free energy change ΔF as:

$$\Delta F = \Delta F_{\text{PEB}} + \Delta F_{\text{EDL}} = \Delta F_{\text{PEB,ent}} + \Delta F_{\text{PEB,VE}} + \Delta F_{\text{PEB,elec}} + \Delta F_{\text{EDL}}, \quad (27)$$

where $\Delta F_{\text{PEB,ent}}$, $\Delta F_{\text{PEB,VE}}$, and $\Delta F_{\text{PEB,elec}}$ are the changes in the free energies of the PE brushes due to entropic, excluded volume, and electrostatic contributions, whereas ΔF_{EDL} is the

free energy change due to the EDL formation caused by the PE charges.

We consider that ψ_0 (which is of the order of $k_{\text{B}}T/e$; $k_{\text{B}}T$ is the thermal energy and e is the electronic charge) is the average of the electrostatic potential that has been triggered around the vicinity of a single grafted PE molecule; accordingly, assuming that the number of chargeable sites of the PE molecule is the same as the number of monomers N we may write $\Delta F_{\text{PEB,elec}}$ for that PE molecule as:

$$\Delta F_{\text{PEB,elec}} \sim eN\psi_0 \sim k_{\text{B}}TN. \quad (28)$$

Our next aim is to provide the scaling relations that will ensure that the height d of the PE brush is the same as that provided by eqn (4); this implies that electrostatic effects of the PE brush are decoupled from the elastic (entropic) and the excluded volume effects of the PE brush, and hence the electrostatic effects have no bearing on the selection of the brush height. It is worthwhile to note here that in this entire scaling calculation, we have not considered ΔF_{EDL} . This can be justified by noting that the EDL is triggered by the PE brushes, and hence their energy contribution must scale as that of the EDL; therefore, in this scaling picture, it suffices to account for $\Delta F_{\text{PEB,elec}}$. As we are interested to identify the de-coupled regime, we consider the ratio of $\Delta F_{\text{PEB,elec}}$ and $\Delta F_{\text{PEB,ent}}$:

$$\frac{\Delta F_{\text{PEB,elec}}}{\Delta F_{\text{PEB,ent}}} \sim \frac{N^2 a^2}{d^2}. \quad (29)$$

Using eqn (4) to replace d in eqn (29), we can write:

$$\frac{\Delta F_{\text{PEB,elec}}}{\Delta F_{\text{PEB,ent}}} \sim \left(\frac{a}{\sigma w}\right)^{2/3}. \quad (30)$$

Considering the Kuhn segment to be cylindrical (of thickness t), so that $w \sim a^2 t$, we may re-write eqn (30) as:

$$\frac{\Delta F_{\text{PEB,elec}}}{\Delta F_{\text{PEB,ent}}} \sim \left(\frac{1}{\sigma a t}\right)^{2/3}. \quad (31)$$

From eqn (31), we can identify that the critical value of the grafting density (σ_{c}) as:

$$\sigma_{\text{c}} \sim \frac{1}{at}. \quad (32)$$

Therefore, for $\sigma_{\text{c}} \sim \sigma$, the PE electrostatic effects become comparable to the PE entropic effects (and therefore, excluded volume effects). On the other hand, when $\sigma_{\text{c}} \ll \sigma$, electrostatic effects are much larger than the other two effects, whereas for $\sigma_{\text{c}} \gg \sigma$, electrostatic effects are much smaller than the other two effects. For these two conditions ($\sigma_{\text{c}} \ll \sigma$ or $\sigma_{\text{c}} \gg \sigma$), we encounter the de-coupled regime; hence d is not dictated by the electrostatic effects.

3.2 Scaling laws for a PE-grafted nanochannel

In case the PEs are grafted on the inner walls of a nanochannel (of half height h), there is a confinement-enforced restriction on the value of the PE layer thickness d . The reason is that in case $d > h$, the grafted PE molecules on the two walls of the nanochannel will interfere with each other undergoing effects

such as compression, interdigitation, *etc.* Here, therefore, we shall only focus on the case where $d < h$. For this condition, we must have [using $w \sim a^2 t$ and assuming that d is dictated by eqn (4)]:

$$N(\sigma a^4 t)^{1/3} < h \quad (33)$$

The corresponding critical $\sigma(\sigma_h)$ should be:

$$\sigma_h \sim \frac{h^3}{N^3 a^4 t}. \quad (34)$$

By definition, for $\sigma < \sigma_h$, $d < h$, whereas for $\sigma > \sigma_h$, $d > h$. Therefore, in order to ensure that PE brushes, grafted in the inner walls of the nanochannel, do not interfere with the brushes on the opposite wall and have their thicknesses independent of the electrostatic effects, one must ensure a $\sigma - N$ combination that simultaneously ensures $\sigma > \sigma^*$, $\sigma < \sigma_h$, and $\sigma \ll \sigma_c$ or $\sigma \gg \sigma_c$.

3.3 EDL electrostatics of the PE-grafted nanochannel in the decoupled regime

In the decoupled regime, the electrostatic effects of the PE charges do not contribute to the selection of the brush height d . Consequently, the EDL electrostatics of the PE brushes are obtained assuming a constant thickness of the PE brushes. Most remarkably, this has been the case in a plethora of investigations (see the review articles^{58–64} and the papers cited in these review articles), although the explicit condition where such a decoupling between the PE electrostatic effects and the PE brush height is possible was not provided. We consider such a de-coupled regime for a nanochannel grafted with PE brushes, quantified by a $\sigma - N$ combination such that $\sigma > \sigma^*$, $\sigma < \sigma_h$, and $\sigma \ll \sigma_c$ or $\sigma \gg \sigma_c$. Such decoupling allows us to write either:

$$\Delta F \approx \Delta F_{\text{PEB,elec}} + \Delta F_{\text{EDL}} \quad [\text{for } \sigma \ll \sigma_c, \sigma > \sigma^*, \sigma < \sigma_h], \quad (35)$$

or

$$\Delta F \gg \Delta F_{\text{PEB,elec}} + \Delta F_{\text{EDL}} \quad [\text{for } \sigma \gg \sigma_c, \sigma > \sigma^*, \sigma < \sigma_h]. \quad (36)$$

Here we propose a model that is valid for both eqn (35) and (36). The nanochannel (of height $2h$) is grafted with PE brushes of thickness d , which is independent of the electrostatic effects. The PE brush is in contact with a symmetric and monovalent electrolyte with a given bulk pH or pOH. The nanochannel is connected by a bulk reservoir; the bulk concentrations of the electrolyte and the hydrogen and hydroxide ions are the concentrations in this bulk reservoir. Under this condition, we may write:

$$\Delta F_{\text{PEB,elec}} + \Delta F_{\text{EDL}} = \int f(\psi, \psi', n_{\pm}, n_{\text{H}^+}, n_{\text{OH}^-}) d^3 \mathbf{r}. \quad (37)$$

Here f is the free energy density, expressed as:^{65,66}

$$\begin{aligned} f = & k_{\text{B}} T \sum_i \left[n_i \left(\ln \left(\frac{n_i}{n_{i,\infty}} \right) - 1 \right) \right] \\ & - \frac{\varepsilon_0 \varepsilon_{\text{r}}}{2} |\nabla \psi|^2 + e\psi \left[\sum_i n_i - \varphi_{\text{PEB}} n_{\text{A}^-} \right] \\ & \text{(for } -h \leq y \leq -h + d \text{) (negative PE layer),} \\ f = & k_{\text{B}} T \sum_i \left[n_i \left(\ln \left(\frac{n_i}{n_{i,\infty}} \right) - 1 \right) \right] \\ & - \frac{\varepsilon_0 \varepsilon_{\text{r}}}{2} |\nabla \psi|^2 + e\psi \left[\sum_i n_i + \varphi_{\text{PEB}} n_{\text{BH}^+} \right] \\ & \text{(for } -h \leq y \leq -h + d \text{) (positive PE layer),} \\ f = & k_{\text{B}} T \sum_i \left[n_i \left(\ln \left(\frac{n_i}{n_{i,\infty}} \right) - 1 \right) \right] \\ & - \frac{\varepsilon_0 \varepsilon_{\text{r}}}{2} |\nabla \psi|^2 + e\psi \left[\sum_i n_i \right] \\ & \text{(for } -h + d \leq y \leq 0 \text{).} \end{aligned} \quad (38)$$

Here ψ is the electrostatic potential, ε_0 is the permittivity of free space, ε_{r} is the relative permittivity of the medium, e is the electronic charge, n_i and $n_{i,\infty}$ are the number density and the bulk number density of the ion of type i ($i = \pm, \text{H}^+, \text{OH}^-$). In addition to the scaling conditions illustrated in eqn (35) and (36), the main assumptions under which eqn (38) is expressed are (a) the electrolyte, hydrogen and hydroxide ions are point charges, (b) the electrostatics are described in a mean field framework disregarding the ion-ion correlations, (c) the solvent acts as a background dielectric without any polarization effects, and (d) the relative permittivity is assumed to be the same both inside and outside the PE brushes. It is worthwhile to note here that for all the three parts of eqn (38), the first to the third terms respectively represent the entropy change due to mixing, the self energy of the EDL electric field, and the electrostatic energy of the PE and ion charges. The most important issue in eqn (38) is the manner in which we express the contribution of the electrostatic energy associated with the PE charges, expressed as $-e\psi\varphi_{\text{PEB}}n_{\text{A}^-}$ (for negatively charged PE brushes) and $e\psi\varphi_{\text{PEB}}n_{\text{BH}^+}$ (for positively charged PE brushes). Here φ_{PEB} represents the dimensionless distribution of the PE chargeable sites;^{65,66} we have shown φ_{PEB} to be cubic in y , expressed as (see Fig. 4):

$$\varphi_{\text{PEB}}(y) = \frac{4}{d^3} (y + h - d)^2 \left(y + h + \frac{d}{2} \right). \quad (39)$$

Significance of this cubic profile will be discussed later. Finally we consider that the PE brushes get charged due to the ionization of the PE chargeable sites. This ionization depends on the local value (within the brush) of the hydrogen and hydroxide concentrations for the negative and the positive PE brushes, respectively. This stems from the fact that for the negative PE brush, the charging occurs on account of an acid-like dissociation, *i.e.*,



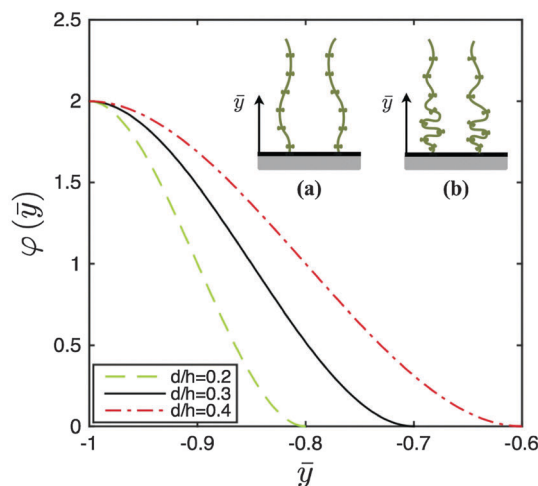
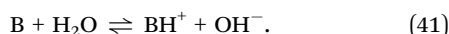


Fig. 4 Cubic profile of the monomer dimensionless number density distribution predicted by Chen and Das.^{65,66} Results are shown for different values of the dimensionless PE layer thickness d/h . In the inset we schematically differentiate between (a) the case of uniform monomer density distribution and (b) the case of cubic monomer distribution. This figure has been adapted with permission from G. Chen and S. Das, Electroosmotic transport in polyelectrolyte-grafted nanochannels with pH-dependent charge density, *J. Appl. Phys.* 2015, **117**, 185304. Copyright (2015), AIP Publishing LLC.

whereas for the positive PE brush, the charging is caused by a base-like dissociation:



Consequently, the number densities of the PE charges, namely n_{A^-} and n_{BH^+} , can be expressed as:

$$n_{A^-} = \frac{K_a' \gamma_a}{K_a' + n_{H^+}}, \quad n_{BH^+} = \frac{K_b' \gamma_b}{K_b' + n_{OH^-}} \quad (42)$$

Here $K_a' = 10^3 N_A K_a$, with K_a being the ionization constant (in units of M) for the dissociation of the acid HA [see eqn (40)] and N_A being the Avogadro number. Also $K_b' = 10^3 N_A K_b$, with K_b being the ionization constant (in units of M) for the ionization of the base B [see eqn (41)]. Finally γ_a and γ_b are the maximum number densities of the PE chargeable sites for the negative and the positive PE brushes, respectively.

The modified equation obtained by using eqn (42) to replace n_{A^-} and n_{BH^+} in eqn (38) will be minimized with respect to ψ , $n_{\pm}$, n_{H^+} , and n_{OH^-} by employing a variational procedure. The interested readers may see Chen and Das⁶⁵ for the detailed derivation; here, we summarize the key equations resulting from this minimization procedure separately for the positive and the negative PE brushes.

Negatively charged PE brushes. Minimization of eqn (38) with respect to ψ yields:

$$\frac{d^2\psi}{dy^2} = \frac{e(n_- - n_+ + n_{OH^-} - n_{H^+} + \varphi_{PEB} n_{A^-})}{\varepsilon_0 \varepsilon_r} \quad (-h \leq y \leq -h + d),$$

$$\frac{d^2\psi}{dy^2} = \frac{e(n_- - n_+ + n_{OH^-} - n_{H^+})}{\varepsilon_0 \varepsilon_r} \quad (-h + d \leq y \leq 0). \quad (43)$$

Minimization of eqn (38) with respect to $n_{\pm}$ yields:

$$n_{\pm} = (n_{\pm,\infty}) \exp\left(\mp \frac{e\psi}{k_B T}\right) \quad (\text{for } -h \leq y \leq 0). \quad (44)$$

Minimization of eqn (38) with respect to n_{OH^-} yields:

$$n_{OH^-} = (n_{OH^-,\infty}) \exp\left(\frac{e\psi}{k_B T}\right) \quad (\text{for } -h \leq y \leq 0). \quad (45)$$

Finally, minimization of eqn (38) with respect to n_{H^+} yields:

$$n_{H^+} = (n_{H^+,\infty}) \exp\left[-\frac{e\psi}{k_B T} \left(1 + \varphi_{PEB} \frac{K_a' \gamma_a}{(K_a' + n_{H^+})^2}\right)\right] \quad (\text{for } -h < y < -h + d), \quad (46)$$

$$n_{H^+} = (n_{H^+,\infty}) \exp\left(-\frac{e\psi}{k_B T}\right) \quad (\text{for } -h + d \leq y \leq 0).$$

We use eqn (42) to replace n_{A^-} in terms of n_{H^+} and use eqn (44) and (45) to replace $n_{\pm}$ and n_{OH^-} in terms of ψ in eqn (43) to obtain:

$$\begin{aligned} \frac{d^2\psi}{dy^2} = & \frac{e}{\varepsilon_0 \varepsilon_r} \left[(n_{-, \infty}) \exp\left(\frac{e\psi}{k_B T}\right) - (n_{+, \infty}) \exp\left(-\frac{e\psi}{k_B T}\right) \right. \\ & \left. + (n_{OH^-, \infty}) \exp\left(\frac{e\psi}{k_B T}\right) - n_{H^+} + \varphi_{PEB} \frac{K_a' \gamma_a}{K_a' + n_{H^+}} \right] \\ & (-h \leq y \leq -h + d), \end{aligned}$$

$$\begin{aligned} \frac{d^2\psi}{dy^2} = & \frac{e}{\varepsilon_0 \varepsilon_r} \left[(n_{-, \infty}) \exp\left(\frac{e\psi}{k_B T}\right) - (n_{+, \infty}) \exp\left(-\frac{e\psi}{k_B T}\right) \right. \\ & \left. + (n_{OH^-, \infty}) \exp\left(\frac{e\psi}{k_B T}\right) - n_{H^+} \right] \quad (-h + d \leq y \leq 0). \end{aligned} \quad (47)$$

We shall need to solve two coupled ordinary differential equations in ψ and n_{H^+} [see eqn (46) and (51)]. The boundary conditions for the potential ψ are well known, namely:

$$\begin{aligned} \left(\frac{d\psi}{dy}\right)_{y=0} &= 0, \quad (\psi)_{y=(-h+d)^+} = (\psi)_{y=(-h+d)^-}, \\ \left(\frac{d\psi}{dy}\right)_{y=(-h+d)^+} &= \left(\frac{d\psi}{dy}\right)_{y=(-h+d)^-}, \quad \left(\frac{d\psi}{dy}\right)_{y=-h} = 0. \end{aligned} \quad (48)$$

Of course the corresponding boundary conditions for n_{H^+} are:

$$\begin{aligned} \left(\frac{dn_{H^+}}{dy}\right)_{y=0} &= 0, \quad (n_{H^+})_{y=(-h+d)^+} = (n_{H^+})_{y=(-h+d)^-}, \\ \left(\frac{dn_{H^+}}{dy}\right)_{y=(-h+d)^+} &= \left(\frac{dn_{H^+}}{dy}\right)_{y=(-h+d)^-}, \quad \left(\frac{dn_{H^+}}{dy}\right)_{y=-h} = 0. \end{aligned} \quad (49)$$

It is easy to find that while eqn (46) ensures that the condition $\left(\frac{dn_{H^+}}{dy}\right)_{y=0} = 0$ results from the conditions $\left(\frac{d\psi}{dy}\right)_{y=0} = 0$, other conditions identified in eqn (48) and (49) can hold

simultaneously only if the function $\varphi_{\text{PEB}}(y)$ obeys the following condition:

$$\begin{aligned} [\varphi_{\text{PEB}}(y)]_{y=-h+d} &= 0, & \left[\frac{d\varphi_{\text{PEB}}(y)}{dy} \right]_{y=-h+d} &= 0, \\ \left[\frac{d\varphi_{\text{PEB}}(y)}{dy} \right]_{y=-h} &= 0. \end{aligned} \quad (50)$$

Additionally, $\varphi_{\text{PEB}}(y)$ must satisfy the condition identified in eqn (7), where “ N ” in eqn (7) should be considered not the total number of monomers in the polymer brush, but rather the total number of chargeable sites in the PE brush. These four conditions can simultaneously be satisfied only if $\varphi_{\text{PEB}}(y)$ is *at least* a cubic function, as expressed in eqn (39). Therefore, we identify the most interesting situation where for a PE brush with pH-dependent charge density, the dimensionless distribution of the chargeable sites must obey a non-unique cubic profile. Fig. 5 shows the variation of the corresponding hydrogen ion number density and the dimensionless electrostatic potential – results demonstrate the distinct impact of the variation of the bulk pH and the $\text{p}K_{\text{a}}$ of the PE acid group whose dissociation leads to PE charging.

Positively charged PE brushes. For the positively charged PE brushes, the same minimization procedure will yield eqn (44) as well as the following set of equations:

$$\frac{d^2\psi}{dy^2} = \frac{e(n_- - n_+ + n_{\text{OH}^-} - n_{\text{H}^+} - \varphi_{\text{PEB}}n_{\text{BH}^+})}{\epsilon_0\epsilon_r}, \quad (-h \leq y \leq -h+d) \quad (51)$$

$$\frac{d^2\psi}{dy^2} = \frac{e(n_- - n_+ + n_{\text{OH}^-} - n_{\text{H}^+})}{\epsilon_0\epsilon_r} \quad (-h+d \leq y \leq 0),$$

$$n_{\text{H}^+} = (n_{\text{H}^+, \infty}) \exp\left(-\frac{e\psi}{k_{\text{B}}T}\right) \quad (\text{for } -h \leq y \leq 0), \quad (52)$$

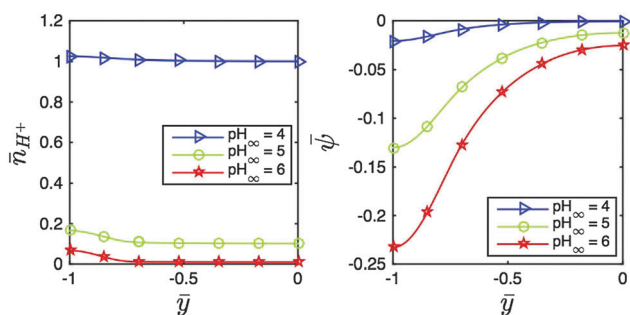


Fig. 5 Variation of (left) the dimensionless hydrogen ion number densities and (right) the dimensionless EDL electrostatic potential for a negatively charged PEL ($\text{p}K_{\text{a}} = 5$) for different values of the bulk pH (or pH_{∞}). Other parameters for the plot are $c_{\infty} = 10^{-4}$ M (bulk concentration of electrolyte ions), $h = 100$ nm (nanochannel height), $\bar{d} = 0.3$ (dimensionless PE layer thickness), and $\gamma_{\text{a}} = 10^{-4}$ M (concentration of PE chargeable sites). This figure probes parameter space different to that probed in our recent paper (ref. 57).

$$n_{\text{OH}^-} = (n_{\text{OH}^-, \infty}) \exp\left[\frac{e\psi}{k_{\text{B}}T} \left(1 + \varphi \frac{K_{\text{b}}'\gamma_{\text{b}}}{(K_{\text{b}}' + n_{\text{OH}^-})^2}\right)\right] \quad (-h < y < -h+d), \quad (53)$$

$$n_{\text{OH}^-} = (n_{\text{OH}^-, \infty}) \exp\left(\frac{e\psi}{k_{\text{B}}T}\right) \quad (-h+d \leq y \leq 0).$$

We use eqn (42) to replace n_{BH^+} in terms of n_{OH^-} and use eqn (44) and (52) to replace $n_{\pm}$ and n_{H^+} in terms of ψ in eqn (43) to obtain:

$$\begin{aligned} \frac{d^2\psi}{dy^2} &= \frac{e}{\epsilon_0\epsilon_r} \left[(n_{-, \infty}) \exp\left(\frac{e\psi}{k_{\text{B}}T}\right) - (n_{+, \infty}) \exp\left(-\frac{e\psi}{k_{\text{B}}T}\right) \right. \\ &\quad \left. + n_{\text{OH}^-} - n_{\text{H}^+, \infty} \exp\left(-\frac{e\psi}{k_{\text{B}}T}\right) - \varphi_{\text{PEB}} \frac{K_{\text{b}}'\gamma_{\text{b}}}{K_{\text{b}}' + n_{\text{OH}^-}} \right] \\ &\quad (-h \leq y \leq -h+d), \\ \frac{d^2\psi}{dy^2} &= \frac{e}{\epsilon_0\epsilon_r} \left[(n_{-, \infty}) \exp\left(\frac{e\psi}{k_{\text{B}}T}\right) - (n_{+, \infty}) \exp\left(-\frac{e\psi}{k_{\text{B}}T}\right) \right. \\ &\quad \left. + n_{\text{OH}^-} - n_{\text{H}^+, \infty} \exp\left(-\frac{e\psi}{k_{\text{B}}T}\right) \right] \\ &\quad (-h+d \leq y \leq 0). \end{aligned} \quad (54)$$

We shall now need to solve two coupled ordinary differential equations in ψ and n_{OH^-} [see eqn (53) and (54)]. The boundary conditions for the potential ψ are expressed in eqn (48), whereas those for n_{OH^-} are expressed as:

$$\begin{aligned} \left(\frac{dn_{\text{OH}^-}}{dy}\right)_{y=0} &= 0, \quad (n_{\text{OH}^-})_{y=(-h+d)^+} = (n_{\text{OH}^-})_{y=(-h+d)^-}, \\ \left(\frac{dn_{\text{OH}^-}}{dy}\right)_{y=(-h+d)^+} &= \left(\frac{dn_{\text{OH}^-}}{dy}\right)_{y=(-h+d)^-}, \\ \left(\frac{dn_{\text{OH}^-}}{dy}\right)_{y=-h} &= 0. \end{aligned} \quad (55)$$

Like the negatively charged PE brush, here too the same cubic profile of $\varphi_{\text{PEB}}(y)$ [see eqn (39)] will ensure that all the conditions of eqn (48) and (55) are simultaneously satisfied. Therefore, we identify that pH-dependent charge density of the positively charged PE brush also ensures that the dimensionless distribution of the chargeable sites must obey a non-unique cubic profile.

3.4 EDL electrostatics of the PE brush in the coupled regime

Quite intuitively, this is the most involved regime where the electrostatic energy contribution of the PE brushes cannot be decoupled from the elastic and the excluded volume energies. The state of the art, in terms of the mathematical representation of this problem, is the paper by Zhulina and Borisov.⁶⁷ In this paper the free energy is expressed as a combination of the free energy expressions used in eqn (5), (37) and (38). However, the major limitation is that the PE charge density despite being assumed to be pH-dependent [identical to that expressed in eqn (38)], the H^+ ion number density is always assumed to be dictated by the Boltzmann distribution, *i.e.*, the second part of eqn (46) both inside and outside the PE brush. This is

inappropriate and stems from the fact that the relationship between the H^+ ion number density and the electrostatic potential ψ is expressed in an *ad hoc* fashion by using Boltzmann distribution. In our recent paper,⁶⁵ we have shown that such “Boltzmann distribution representation” of the H^+ ion number densities leads to unphysical discontinuities in the value and the gradient of the H^+ ion number densities at the PE-brush–electrolyte interface, and we proposed a cubic distribution of the PE chargeable sites within the brush depth to account for this discontinuity. Of course, our derivation, as has been discussed in the previous subsection, was under the assumption that the PE electrostatic effects are decoupled from the elastic and the excluded volume effects of the PE brushes. This in turn will imply that the dimensionless distribution of the PE chargeable sites may not be related to the dimensionless distribution of the monomers. In this coupled regime, such mutual independence of the PE chargeable sites and the monomers is not possible. Our analysis would need to start from the full free energy expression and obtain the equilibrium conditions from variational minimization. This procedure should ideally yield a mutually related distribution of the monomers and the PE chargeable sites in a manner such that energies are minimized and at the same time there is no unphysical discontinuities of the value and gradient of the hydrogen ion number densities at the PE-brush–electrolyte interface. This calculation is yet to be done, and represents a major open question in the mathematical description of the PE brushes with pH-dependent charge density.

3.5 Other important descriptions of the mathematics of the PE brushes

The EDL electrostatics of the PE brushes described in Section 3.3 assumes a mean-field framework with the ions represented as point charges. Recently, we have developed models that capture the PE EDL electrostatics by considering finite ion sizes; though still in a mean-field framework, such considerations reveal crucial finite-ion-size-dependence of EDL electrostatic potential and the resulting disjoining pressure between two soft charged plates (see Fig. 6).^{68,69} The results clearly indicate that the finite ion size effect (quantified through the parameter $\nu = 2n_\infty a^3$, where a is the ion size) leads to an enhancement of the interaction force (characterized by the disjoining pressure) between two soft charged plates. Of course, these calculations considered that the PE electrostatic contributions are decoupled from the elastic and the excluded volume contributions. Some of the related open research questions are to probe the effects of non-mean-field effects such as ionic correlation in computing the PE brush EDL electrostatics for cases where the electrostatic effects may or may not be decoupled from the elastic and excluded volume effects.

The next important issue is the effect of the solid boundary on which the PE brushes are grafted on the PE brush EDL electrostatics. Most of the studies on PE brush electrostatics have considered this solid surface as chargeless.^{58–66} While this condition appropriately represents the case of a metallic solid, it is inappropriate to represent the biological soft systems (*e.g.*, cells, bacterial, viruses, *etc.*) whose surfaces can often be approximated as PE-grafted interfaces. For example, Dingari and Buie described bacterial electrostatics and electrokinetics by considering this solid

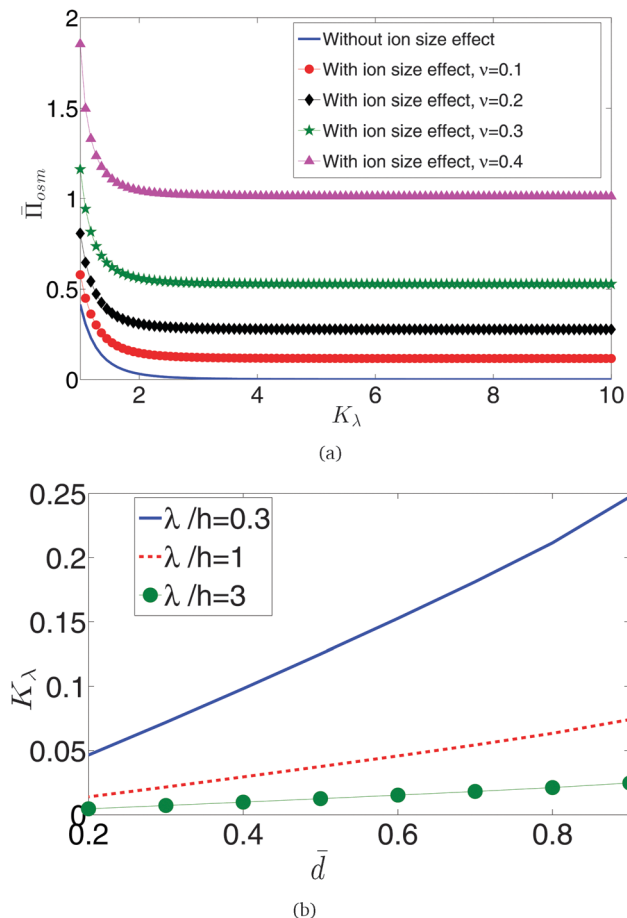


Fig. 6 (a) Variation of dimensionless osmotic pressure $\bar{\Pi}_{osm} = \Pi / (2n_\infty k_B T)$ (where Π_{osm} is the osmotic pressure) quantifying the interaction force between two soft charged plates with $K_\lambda = \lambda_{PElayer} / \lambda$ (where $\lambda_{PElayer}$ is the thickness of the equivalent EDL thickness of the PE layer and λ is the electrolyte EDL thickness) for the case with finite ion size effects denoted in terms of the Steric parameter ν . (b) Variation of K_λ with dimensionless PE layer thickness ($\bar{d} = d/h$) for different values of dimensionless EDL thickness (λ/h). Both (a) and (b) are reproduced from ref. 69 with permission from The Royal Society of Chemistry.

as a finite dielectric solid.⁷⁰ Similarly, Phan *et al.*⁷¹ and McDaniel *et al.*⁷² described the electrostatics of the MS2 bacteriophage virus by considering the virus as a dielectric solid with a charged core.

Other relevant theoretical studies of electrostatics of PE brushes include description of electrostatics of poly-zwitterion brushes by Milne *et al.*,⁷³ delineation of the effect of poor solvent by Ross and Pincus⁷⁴ and Borisov *et al.*,⁷⁵ probing of interfacial instability of the end-charged PE brushes,⁷⁶ quantification of ionic thickness dependence of PE brushes,⁷⁷ quantification of entropy-driven collective interactions of DNA brushes,⁷⁸ and many more.

4 Transport at interfaces grafted with PE brushes

4.1 Liquid transport at interfaces grafted with PE brushes

Understanding liquid transport at interfaces grafted with polymer and PE brushes is central to a large number of natural phenomena

and technological applications. Some of them are regulating fluid flow in polymer-brush or PE-brush grafted micro-nanochannels leading to designing of nano valves,^{79,80} nanochannel streaming potential induced electrochemomechanical energy conversion,^{81,82} suppressing electroosmotic transport for enhancing the efficiency of processes depending on the detection and measurement of nanochannel ionic current,⁶⁶ etc.

The discussions that follow in this subsection as well as the following subsections are based on the key assumption that there is no change in the morphology of the PE brushes due to the shear imparted by the flow field or the force imparted by external fields (e.g., external electric field). Such an assumption necessitates that the shear force imparted by the background flow should be much smaller than the elastic restoring force associated with the deforming PE brush. Of course there are studies that have attempted to incorporate the effect of flow-induced deformation of the PE or polymer brushes self-consistently while calculating the liquid transport at PE or polymer-grafted interfaces.^{83,84} Challenges are that such coupling often makes the calculations too complicated, thereby forbidding analysis of several other important effects related to polymer and PE brushes (e.g., effects induced by the charges of the PEs). The only remaining option is the employment of computationally expensive atomistic and molecular simulation models (discussed in details in Section 5). Also in most of the realistic settings (e.g., transport in PE-grafted nanochannels^{66,82} or electrophoresis of PE-grafted nanoparticles^{59–61}), the deforming forces induced by the fluid shear or the external field are too small to cause any significant deformation of the PE brushes, thereby ensuring that the PE brush configuration may be assumed to be independent of the background flow or field effects.

The most fundamental issue of modelling liquid transport in PE grafted channels is to quantify the drag force imparted on the liquid transport by the PE brushes. A fully-developed, steady-state, one-dimensional pressure-driven flow in the PE or polymer grafted channel (or micro-nanochannel) can be described as:^{81,85}

$$\eta \frac{d^2 u}{dy^2} - \frac{dp}{dx} - \mu_c u = 0, \quad (56)$$

where u is the axial velocity field, dp/dx is the constant axial pressure gradient, η is the dynamic viscosity of the liquid, and μ_c is the per unit volume drag coefficient. This equation is equivalent to the Brinkmann equation in a porous medium. μ_c can be interpreted in terms of the length scale across which the background flow is screened from the flow inside the polymer coil with the polymer being in a semi-dilute solution.^{86,87} The present case of grafted polymer or PE brushes is a special case, where the polymer average velocity is always much smaller than the background flow velocity. For such a case, $\mu_c \sim K_{sc}^2$, where K_{sc}^{-1} is the screening length. Since $K_{sc}^{-1} \sim \varphi^{-1}$ (where φ is the dimensionless monomer distribution),⁸⁸ we can write $\mu_c \sim \varphi^2$. The prefactor connecting μ_c and φ is η/a^2 , so that:

$$\mu_c \approx \frac{\eta}{a^2} \frac{1}{\varphi^2}. \quad (57)$$

Analytical and semi-analytical solutions of the velocity profile have been proposed for the simplistic cases where the monomer density distribution has been assumed to obey parabolic profiles and the flow is assumed not to disturb the monomer density distributions.⁸⁵ A more rigorous treatment, considering φ to be governed by eqn (21), has also been done in recent calculations.⁸⁰

It is also possible to trigger an electroosmotic transport in PE grafted micro-nanochannels by employing external electric field (E_x). For such an electroosmotic transport, in addition to accounting for the appropriate drag force, one must also account for the interaction of the PE-EDL charge density and the applied electric field. The fully-developed, steady-state, one-dimensional velocity field is governed by the following equation:

$$\eta \frac{d^2 u}{dy^2} + e E_x \sum_i (z_i n_i) - \mu_c u = 0, \quad (58)$$

where z_i and n_i are the valence and the number densities of the mobile ions within the PE-EDL. Of course, $\mu_c \sim \varphi^2$ must bear the signature of the corresponding monomer density distribution. Therefore, for the case of PE brushes in the decoupled regime (i.e., where the PE electrostatic effects are decoupled from the elastic and the excluded volume effects), the cubic variation in the number density of the PE chargeable sites must be reflected in the corresponding variation of the drag coefficient. Fig. 7 shows our recent results⁶⁶ on the electroosmotic velocity profile for nanochannels grafted with PE brushes having pH-dependent charge density. Results clearly demonstrate the manner in which the net flow is substantially reduced in PE-grafted nanochannels.

Electroosmotic transport in a micro-nanochannel may also occur in the presence of a pure pressure-driven transport. This is the case of the electroosmotic transport in presence of the streaming electric field induced by the pressure-driven downstream migration of the mobile counterions within the EDL.^{89–92} This streaming electric field has been known to generate electrical energy at the nanoscale, and this mechanism has been identified as the Electrochemomechanical Energy Conversion.^{93,94} Moreover, this streaming electric field driven electroosmotic transport retards the pressure-driven transport. As a consequence the net flow rate gets reduced, which is modelled by attributing the flow with an enhanced viscosity – this effect is known as the Electroviscous Effect.⁸⁹ Several researchers including us developed models for quantifying the streaming electric field in PE grafted micro-nanochannels (often identified as “soft” nanochannels).^{81,82,95–105} The basic equation governing the flow field reads:

$$\eta \frac{d^2 u}{dy^2} - \frac{dp}{dx} + e E_s \sum_i (z_i n_i) - \mu_c u = 0, \quad (59)$$

where E_s is the streaming electric field induced due to the pressure-driven transport. E_s is not known *a priori* and should be obtained by equating the net current (sum of the streaming and the conduction currents) in the micro-nanochannel to zero.^{89,90,92} We established that for cases where the EDL electrostatic potential

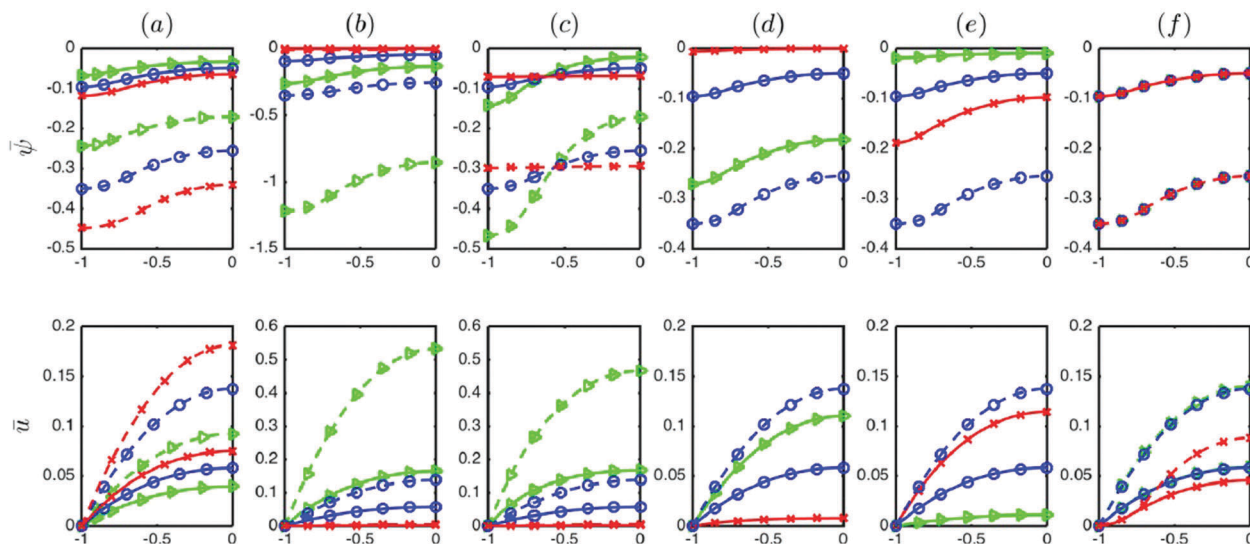


Fig. 7 Transverse variation of $\bar{\psi} = e\psi/(k_B T)$ (dimensionless electrostatic potential) (top panel) and $\bar{u}$ (the dimensionless electroosmotic velocity profile, see Chen and Das⁶⁶ for definition) (bottom panel). The x-axis in all the plots is the non-dimensional distance along the channel bottom height. Dimensionless parameters dictating $\bar{\psi}$ are $\bar{d} = d/h$, K_λ (see the caption of Fig. 7 for definition), $\bar{n}_{H^+, \infty} = n_{H^+, \infty}/n_\infty$, $\bar{\lambda} = \lambda/h$, $\bar{K}_a' = N_A 10^{3-pK_a}$ (N_A is the Avogadro number). On the other hand, $\bar{u}$ is dictated by all these parameters as well as the parameter $\bar{\alpha} = h\alpha = h\sqrt{\mu_c/\eta}$. For all the figures we maintain the five dimensionless parameters (K_λ , $\bar{n}_{H^+, \infty}$, $\bar{\lambda}$, $\bar{K}_a'$, $\bar{\alpha}$) equal to 1 and $\bar{d} = 0.3$, except for (a) $\bar{d} = 0.2$ (triangular markers), $\bar{d} = 0.3$ (circular markers), $\bar{d} = 0.4$ (cross markers), (b) $K_\lambda = 0.5$ (triangular markers), $K_\lambda = 1$ (circular markers), $K_\lambda = 5$ (cross markers), (c) $\bar{\lambda} = 0.5$ (triangular markers), $\bar{\lambda} = 1$ (circular markers), $\bar{\lambda} = 5$ (cross markers) (d) $\bar{n}_{H^+, \infty} = 0.1$ (triangular markers), $\bar{n}_{H^+, \infty} = 1$ (circular markers), $\bar{n}_{H^+, \infty} = 10$ (cross markers), (e) $\bar{K}_a' = 0.1$ (triangular markers), $\bar{K}_a' = 1$ (circular markers), $\bar{K}_a' = 10$ (cross markers), and (f) $\bar{\alpha} = 0.1$ (triangular markers), $\bar{\alpha} = 1$ (circular markers), $\bar{\alpha} = 10$ (cross markers). In (f), we show that $\bar{\alpha}$ affects only $\bar{u}$ and not $\bar{\psi}$. For all the figures, the case with pH effects is represented by solid lines, whereas the case without pH effects is represented by dashed lines. For other parameters, kindly refer to Chen and Das.⁶⁶ This figure has been reproduced with permission from G. Chen and S. Das, Electroosmotic transport in polyelectrolyte-grafted nanochannels with pH-dependent charge density, *J. Appl. Phys.* 2015, **117**, 185304. Copyright (2015), AIP Publishing LLC.

and electroosmotic velocity field can be expressed analytically, E_s can be obtained by expressing it explicitly in terms of the EDL electrostatic potential and hence determined by simple numerical integration.⁸¹ On the contrary, for cases where the EDL electrostatic potential and the consequent electroosmotic transport are not analytically expressible (e.g., for cases where the electrostatic potential is large enough to invalidate the use of Debye-Hückel linearization or for cases where the PE charge density is pH-dependent^{65,66,72}), E_s needs to be obtained by solving an integro-differential equation developed in Chen and Das.⁸² Fig. 8(a and b) show the streaming potential and the electroviscous effects for this case where the EDL electrostatic potential at the PE-grafted interface is described using the non-Debye-Hückel approach.

In most of these models on electroosmotic transport in PE grafted micro-nanochannels, the EDL electrostatic potential and the ion distribution have been assumed to be unaffected by the flow field. An acceptable justification is that the time scale of formation of the EDL $\tau_{EDL} \sim \lambda^2/D_{ion}$ (where D_{ion} is the diffusion coefficient of the ions and λ is the EDL thickness), and therefore $\tau_{EDL} \sim 10^{-6}$ – 10^{-8} s, which is much smaller than the timescale associated with the electroosmotic transport. Of course, there are studies where this simplification has been discarded and the electroosmotic transport in a “soft” nanochannel has been obtained by solving coupled fluid flow and ionic transport equations.⁷³

4.2 Ionic transport at interfaces grafted with PE brushes

Calculation of nanochannel electroosmotic transport typically neglects the ion transport (and the resulting ionic current).

However, the truth is that this ionic current is often a more important parameter in the context of nanochannel transport. This stems from the fact that detection, regulation, and manipulation of this ionic current is central to the majority of applications involving nanofluidic transport.

Detection and manipulation of ionic current is also central to the functioning of PE grafted nanochannels. In fact, this PE grafting renders a desired “smartness” or functionality to the nanochannels that allows them to better control and manipulate the ionic current leading to a plethora of applications. This ionic current (per unit width), triggered in the presence of an external axial electric field E , can be expressed as:

$$i_{ion} = eE \sum_i \frac{z_i}{|z_i|} \Xi_i n_i, \quad (60)$$

where Ξ_i is the electrophoretic mobility of ionic species i . In our recent study,⁶⁶ we showed that for a PE-grafted nanochannel the ionic current can be utilized for different purposes (e.g., current rectification and gating,^{106–108} ion and biosensing,^{109–112} etc.) more effectively, stemming from a massive suppression of the corresponding electroosmotic transport. In another recent study,⁵⁷ we demonstrated a possible inversion in the ionic current for certain choices of parameters: this inversion, which characterizes a “coion-dominated” ionic current, occurs on account of the bulk ionic current dominating the EDL ionic current.

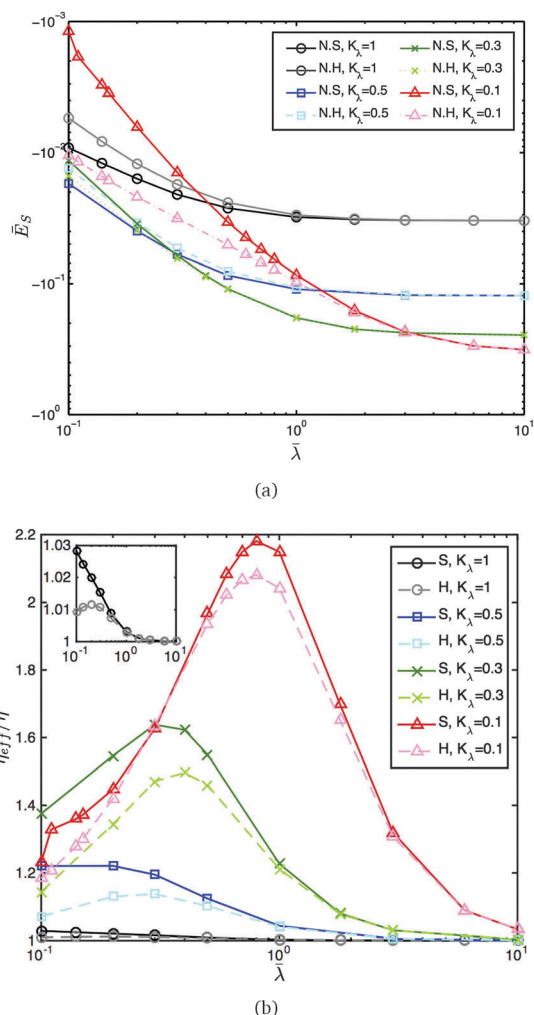


Fig. 8 (a) Variation of the dimensionless streaming potential $\bar{E}_S$ (see Chen and Das⁸² for details on this non-dimensionalization procedure) with dimensionless EDL thickness $\bar{\lambda} = \lambda/h$ for the soft nanochannel (or N.S.) and the rigid nanochannel (N.H.) for different values of K_λ (defined in the caption of Fig. 6). (b) Variation of the viscoisty ratio (quantifying the electroviscous effect) with dimensionless EDL thickness for the soft nanochannel (denoted as S) and the rigid nanochannel (denoted as H) for different values of K_λ . For both (a) and (b), the hard nanochannel (the nanochannel without the PE layer) is assumed to have the same zeta potential at the solid-PE-layer interface as that of the soft nanochannel (or the nanochannel with the PE layer). For other parameters of these plots please refer to Chen and Das.⁸² This figure is reprinted from G. Chen and S. Das, Streaming potential and electroviscous effects in soft nanochannels beyond Debye–Hückel linearization, *J. Colloid Interface Sci.*, 2015, **445**, 357–363, Copyright, with permission from Elsevier.

4.3 Electrophoresis of micro-nano particles grafted with PE brushes

Electrophoresis of moieties grafted with PE brush – examples can be non-biological systems such as PE-grafted nanoparticles or biological species such as cells, bacteria, and viruses – requires understanding of both ion and liquid transport at PE-grafted interfaces. Here we provide the key mathematical steps for quantifying the electrophoresis of PE-grafted micro-nanoparticles; interested readers are directed to a large volume of papers by Ohshima, Duval, and co-workers.^{58,61,113–124}

We start by considering a spherical uncharged particle of radius a (of course, this “ a ” is different from the polymer Kuhn length “ a ” defined in Section 2), grafted with a PE brush of thickness d ($b = a + d$). Throughout the calculation, it is assumed that the PE electrostatic effects are decoupled from the corresponding elastic and the excluded volume effects; in other words, this allows us to assume a constant d . The particle is assumed to move with a velocity $\mathbf{U}$ in the presence of an applied electric field $\mathbf{E}$. The main aim of this proposed derivation is to obtain the ratio $|\mathbf{U}|/|\mathbf{E}|$, which is the electrophoretic mobility of the particles. In order to obtain this ratio, we shall first need to obtain the fluid velocity field $\mathbf{u}$ governed by the following sets of equations and boundary conditions:

$$\eta \nabla \times \nabla \times \mathbf{u} + \nabla p + e \sum_i [z_i n_i(\mathbf{r})] \nabla \psi = \mathbf{0} \quad r > b,$$

$$\eta \nabla \times \nabla \times \mathbf{u} + \mu_c \mathbf{u} + \nabla p + e \sum_i [z_i n_i(\mathbf{r})] \nabla \psi = \mathbf{0} \quad a < r < b, \quad (61)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (62)$$

$$\mathbf{u}_{r=a} = (u_r \mathbf{e}_r + u_\theta \mathbf{e}_\theta + 0 \mathbf{e}_\phi)_{r=a} = \mathbf{0},$$

$$\mathbf{u}_{r \rightarrow \infty} \rightarrow -\mathbf{U} = (-U \cos \theta \mathbf{e}_r + U \sin \theta \mathbf{e}_\theta + 0 \mathbf{e}_\phi),$$

$$(u_r)_{r=b^-} = (u_r)_{r=b^+}, (u_\theta)_{r=b^-} = (u_\theta)_{r=b^+}. \quad (63)$$

Like any electrokinetic transport, here too the calculations rely on the appropriate description of the number densities n_i of the mobile ions in the EDL. Since here the electrostatic effects are assumed to be decoupled from the corresponding elastic and the excluded volume effects of the PE brushes, we can use the number density distributions derived from free energy minimization approach elucidated in Section 3.2. For the present case, we consider the simplistic case where the charge density of the PE brushes is constant and pH-independent. For such cases, we only consider the electrolyte ions (*i.e.*, $i = \pm$) and not the H^+ and OH^- ions; accordingly for a positively charged PE brush we can obtain

$$\nabla^2 \psi = -\frac{e[n_+(\mathbf{r}) - n_-(\mathbf{r})]}{\epsilon_0 \epsilon_r} \quad r > b,$$

$$\nabla^2 \psi = -\frac{e[n_+(\mathbf{r}) - n_-(\mathbf{r})] + \gamma_b}{\epsilon_0 \epsilon_r} \quad a < r < b, \quad (64)$$

$$\mu_\pm(\mathbf{r}) = \mu_{0,\pm} \pm e\psi(\mathbf{r}) + k_B T \ln[n_\pm(\mathbf{r})], \quad (65)$$

$$\mathbf{v}_\pm(\mathbf{r}) = \mathbf{u} - \frac{1}{\xi_\pm} \nabla \mu_\pm, \quad (66)$$

$$\nabla \cdot (n_\pm \mathbf{v}_\pm) = 0, \quad (67)$$

$$(\mathbf{v}_\pm \cdot \hat{\mathbf{n}})_{r=a} = 0. \quad (68)$$

In the above equations, γ_b is the number density of the positively charged PE layer, $\xi_\pm$ is the drag coefficient of the ions, $\mathbf{v}_\pm$ is the ionic velocity vector, $\mathbf{n}$ is the unit vector along the normal at any point on the spherical particle, and $\mu_\pm$ is the electrochemical potential of the ions.

We consider weak applied electric field; hence we can express:

$$n_{\pm}(\mathbf{r}) = n_{\pm}^{(0)}(r) + \delta n_{\pm}(\mathbf{r}), \quad (69)$$

$$\psi(\mathbf{r}) = \psi^{(0)}(r) + \delta\psi(\mathbf{r}), \quad (70)$$

$$\mu_{\pm}(\mathbf{r}) = \mu_{\pm}^{(0)} + \delta\mu_{\pm}(\mathbf{r}). \quad (71)$$

Please note that quantities with superscripts (0) refer to those at equilibrium, *i.e.*, in the absence of the external electric field E . Also note that $n_{\pm}^{(0)}$ and $\psi^{(0)}$ are functions of only r , whereas the electrochemical potential $\mu_{\pm}^{(0)}$ is constant and independent of r .

One can employ a perturbation approach, which eventually provides an expression for the electrophoretic mobility (Ξ_{soft}) of the PE-grafted particle (for detailed derivation, kindly refer to Ohshima¹¹⁵):

$$\begin{aligned} \Xi_{\text{soft}} = & \frac{b^2}{9} \int_b^{\infty} \left[3 \left(1 - \frac{r^2}{b^2} \right) - \frac{2L_2}{L_1} \left(1 - \frac{r^3}{b^3} \right) \right] G(r) dr \\ & + \frac{2L_3}{3\alpha^2 L_1} \int_b^{\infty} \left(1 + \frac{r^3}{2b^3} \right) G(r) dr \\ & - \frac{2}{3\alpha^2} \int_a^b \left[1 - \frac{3a}{2\alpha^2 b^3 L_1} (L_3 + L_4 \alpha r) \cosh[\alpha(r-a)] \right] G(r) dr \\ & + \frac{2}{3\alpha^2} \int_a^b [(L_4 + L_3 \alpha r) \sinh[\alpha(r-a)]] G(r) dr, \end{aligned} \quad (72)$$

where

$$\begin{aligned} L_1 = & \left(1 + \frac{a^3}{2b^3} + \frac{3a}{2\alpha^2 b^3} - \frac{3a^2}{2\alpha^2 b^4} \right) \cosh[\alpha(b-a)] \\ & - \left(1 - \frac{3a^2}{2b^2} + \frac{a^3}{2b^3} + \frac{3a}{2\alpha^2 b^3} \right) \frac{\sinh[\alpha(b-a)]}{\alpha b}, \end{aligned} \quad (73)$$

$$\begin{aligned} L_2 = & \left(1 + \frac{a^3}{2b^3} + \frac{3a}{2\alpha^2 b^3} \right) \cosh[\alpha(b-a)] \\ & + \frac{3a^2}{2b^2} \frac{\sinh[\alpha(b-a)]}{\alpha b} - \frac{3a}{2\alpha^2 b^3}, \end{aligned} \quad (74)$$

$$L_3 = \cosh[\alpha(b-a)] - \frac{\sinh[\alpha(b-a)]}{\alpha b} - \frac{a}{b}, \quad (75)$$

$$L_4 = \sinh[\alpha(b-a)] - \frac{\cosh[\alpha(b-a)]}{\alpha b} - \frac{\alpha a^2}{3b} + \frac{2\alpha b^2}{3a} + \frac{1}{\alpha b}, \quad (76)$$

$$\alpha = (\mu_c/\eta)^{1/2}, \quad (77)$$

$$G(r) = -\frac{en_{\infty}}{\eta r} \left(\frac{d\bar{\psi}^0}{dr} \right) [\exp(-\bar{\psi}^0)\phi_+ + \exp(\bar{\psi}^0)\phi_-], \quad (78)$$

$$\begin{aligned} \phi_{\pm}(r) = & \mp \frac{1}{3} \left(r + \frac{a^3}{2r^2} \right) \int_a^{\infty} \frac{d\bar{\psi}^0}{dx} \left[\frac{d\phi_{\mp}(x)}{dx} \mp \frac{2\alpha h_{\text{ep}}(x)}{e} \frac{1}{x} \right] dx \\ & \pm \int_a^r \left(r - \frac{x^3}{r^2} \right) \frac{d\bar{\psi}^0}{dx} \left[\frac{d\phi_{\mp}(x)}{dx} \mp \frac{2\alpha h_{\text{ep}}(x)}{e} \frac{1}{x} \right] dx + r + \frac{a^3}{2r^2}, \end{aligned} \quad (79)$$

and

$$\begin{aligned} h_{\text{ep}}(r) = & \frac{U}{2E} r + \frac{D_1}{r^2} + D_2 + D_3 r^2 + \frac{r^3}{30} \int_{\infty}^r G(x) dx \\ & - \frac{r}{6} \int_{\infty}^r x^2 G(x) dx + \frac{1}{6} \int_{\infty}^r x^3 G(x) dx \\ & - \frac{1}{30r^2} \int_{\infty}^r x^5 G(x) dx \quad r > b, \\ h_{\text{ep}}(r) = & \frac{1}{3\alpha^2} \left[-r \int_b^r G(x) dx + \frac{1}{r^2} \int_b^r x^3 G(x) dx \right] - \frac{C_1 r}{\alpha^2} - \frac{C_2}{\alpha^2 r^2} \\ & + \frac{1}{\alpha^3} \int_b^r \left[\left(\frac{x}{r} - \frac{1}{\alpha^2 r^2} \right) \sinh[\alpha(r-x)] - \left(\frac{x}{\alpha r^2} - \frac{1}{\alpha r} \right) \right. \\ & \times \cosh[\alpha(r-x)] \left. \right] G(x) dx \\ & + C_3 \left[\frac{\alpha \cosh[\alpha(r-b)]}{r} - \frac{\sinh[\alpha(r-b)]}{r^2} \right] \\ & + C_4 \left[\frac{\alpha \sinh[\alpha(r-b)]}{r} - \frac{\cosh[\alpha(r-b)]}{r^2} \right], \quad a < r < b. \end{aligned} \quad (81)$$

There have been several simplifications of eqn (72) for obtaining the “soft” particle electrophoretic mobilities under different limiting conditions.⁶¹ Dingari and Buie⁷⁰ provided one of the rare studies where the electrophoretic mobility was computed considering pH-dependent charge density of the grafted PE brushes. However, their model also suffers from the same limitations as the other studies probing the pH-dependent PE brushes, namely consideration of Boltzmann distribution for describing the hydrogen ion number densities inside the PE brushes. Therefore, an appropriate consideration should describe the electrophoresis in the framework illustrated in Section 3.3 (*i.e.*, the framework that considers cubic distribution of the monomer profile^{65,66}). This is a major open problem in the mathematical analysis of “soft” particle electrophoresis. The other open mathematical problem in this regard is the computation of the electrophoretic mobility of the “soft” particles for the cases where the electrostatic effects of the PE brushes cannot be decoupled from the corresponding elastic and the excluded volume effects.

5 Atomistic and mesoscopic simulations of PE grafted interfaces

Continuum scale calculations for polymer and PE brushes in the presence of a solvent (described in Sections 2–4) may not always be able to capture the solvent–monomer interactions in an accurate fashion. For example, during the swelling or shrinking of a polymer or a PE brush, there may be a possible re-orientation of the solvent molecules, which in turn will change the configuration of the brush. This makes the problem a coupled one, beyond the tractability of the continuum models. For such cases, the problem has been probed by using

Molecular Dynamics (MD) simulation techniques. Below we briefly discuss some of the important studies of MD simulation of polymer and PE brushes.

Initially PBs were modelled as chains of monomers, where the individual monomer–monomer interactions were considered, but the chain repelling effects and the hydrodynamic effects were neglected. The motion was described using the Langevin equation:¹²⁵

$$\ddot{\mathbf{r}}_i = -\nabla U_i - \Gamma \dot{\mathbf{r}}_i + \mathbf{W}_i(t) \quad (83)$$

where, $\mathbf{r}_i$ is the position vector of monomer i , Γ is the monomer–solvent friction and $\mathbf{W}_i(t)$ is the random force vector given through a Gaussian white noise. The potential U_i was divided into a repulsive and an attractive component. The purely repulsive part was expressed as:

$$U^r = 4\varepsilon_{\text{LJ}} \left[\left(\frac{\sigma_{\text{LJ}}}{r} \right)^{12} - \left(\frac{\sigma_{\text{LJ}}}{r} \right)^6 + \frac{1}{4} \right], \quad (84)$$

where ε_{LJ} and σ_{LJ} are the Lennard Jones interaction parameters between the interacting monomers. The repulsive component was truncated beyond $\sigma_{\text{LJ}}^{1/6}$. The attractive component, on the other hand, was expressed as:

$$U^a = -0.5kR_0^2 \ln(1 - (r/R_0)^2) \quad (85)$$

The attractive potential becomes zero beyond $R_0 = 1.5\sigma_{\text{LJ}}$. Here the constant $k = 30\varepsilon_{\text{LJ}}/\sigma_{\text{LJ}}^2$. This model never gives the self-entanglement dynamics, making it attractive for simulating PBs, where self-entanglement is typically absent. Kremer and co-workers used the realistic values for PDMS brushes and validated this model by using it to study the crossover from the Rouse to the reptation dynamics.¹²⁶

A more rigorous model to study the PBs is the bead-spring model. Here the monomers are considered to be the beads having a specific mass concentration, and connected to each other by springs. Nature of the springs dictates the nature of flexibility of the polymers. Grest and co-workers used this model for MD simulations of polymers¹²⁷ as well as to obtain the MD results of flow-induced vertical deformation of a grafted PB system¹²⁸ that matched well with the corresponding experimental results.¹²⁹ Therefore, this model was shown to successfully capture the hydrodynamic effects that was not possible using the model described above. It was further shown that this model, when used with a truncated Lennard–Jones 6–12 potential, can also describe flow-induced deformation of PBs in a horizontal channel.¹³⁰ Moreover, this model yielded a parabolic form of the monomer density profile, which matched the prediction of the self-consistent field theory (SCFT).¹³⁰ Finally, this model described the interpenetration behaviour of the overlapping brushes,¹²⁸ and the results matched well with the scaling predictions.¹³¹ The fundamental basis of this bead-spring model for MD simulation of PBs lies upon the fact that for many properties of the tethered chains, it is possible to use a coarse-grained model instead of a very detailed atomistic one. This stems from the fact that the intrinsic length and the time scales associated with the system were much larger than that of the monomer–monomer interactions.

The polymer reference interaction site model (PRISM),¹³² which is a more advanced model for studying the behaviour of PBs, was proposed on the basis of the integral equation ideas of monoatomic liquids. The starting point of the calculations used in this model is the radial distribution function obtained from the Ornstein–Zernike equation.¹³² The direct correlation function is deduced from the Percus–Yevick approximation. The intramolecular functions and the intermolecular radial distribution functions in the case of flexible polymers depend on each other – therefore they need to be determined in a self-consistent fashion through rigorous calculations.^{133–135} But for the case of a single component polymer melt, one can avoid this self-consistent calculation by nominally using the Flory ideality hypothesis.¹³² PRISM was also used to study the behaviour of semi-flexible PBs.¹³² The “semi-flexibility” was introduced by considering that the monomers were connected by harmonic springs and the probability density between two intramolecular sites was considered to be Gaussian in nature. For a semi flexible chain the intermolecular distribution function was approximated by the Koyama distribution. The PRISM predictions for the behaviour of homopolymers were compared with the MD results obtained using the models described in the previous two paragraphs¹³⁶ – this comparison established that the results from the PRISM showed a better match with the experiments and SCFT results. Stevenson *et al.*¹³⁶ proposed a modified version of the PRISM, where the monomers were represented as Lennard-Jones soft core entities – the corresponding radial distribution functions were developed using the Weeks–Chandler–Andersen model.¹³⁷ Stevenson *et al.*¹³⁶ showed that this modified PRISM can be successfully applied for simulating PBs in various blends, with the simulation results showing excellent match with the results obtained from the integral equations. In a separate paper, Stevenson *et al.*¹³⁸ included effects such as size correction in this modified PRISM.

PE brushes were simulated by considering the bead-spring model (see above) using the FENE (Finite Extendable Nonlinear Elastic) potential with spring constant $k = 7\varepsilon_{\text{LJ}}/\sigma_{\text{LJ}}^2$ and maximum length as $R_0 = 2\sigma_{\text{LJ}}$.^{139,140} This model considers the PEs to be freely combined in nature. The initial difficulties associated with including charges stemmed from the fact that both the charged monomers as well as the counterions needed to be considered simultaneously. Hence wormlike polymers with a persistence length, considered as a sum of the intrinsic persistence length and the electrostatic persistence length, were used. The simulations were performed in a good solvent and revealed a regime where the electrostatic repulsions became less important. Such a regime had not been predicted by experiments, that had been typically performed in poor solvents. In a subsequent study, monovalent counterions were specially added to each of the chains.¹⁴¹ For this study the value of the energy parameter was set to be $0.5k_{\text{B}}T$, which enforced that for a highly charged chain in a near Θ condition, the size of the thermal blob became much larger than the electrostatic blob. In order to account for the purely repulsive volume exclusion effects, a shifted LJ potential was considered. The most important part in these simulations was that the solvent molecules

were not modelled explicitly.^{141,142} A more rigorous approach was the consideration of a dielectric medium in which all the charged particles interacted *via* an unscreened Coulomb potential.¹⁴³ A smoothed particle mesh Ewald algorithm was used for the implementation of the periodic boundary conditions. The Langevin equation was used for studying the motion of each particle. It was also found out that a Nose–Hoover thermostat causes a faster decay of the time correlation functions than the Langevin thermostat. Very recently PEs have been simulated in a capacitor geometry with opposite charged PEs on two capacitor surfaces.¹⁴⁴ The simulations were carried out for a coarse grained system in an implicit solvent. The diameter of the counterions was considered as half the unit length of the beads in order to enforce the condition that the counterions are very small. The Coulomb potentials were used without truncation using a pppm-algorithm.

Study reported by Grest was one of the pioneering studies on the MD simulation of PBs in the presence of a flow.¹³⁰ Grest reported that the drag force induced by the PB layer is smaller than that predicted by continuum theory. Electrokinetic and electroosmotic transports through the PE-grafted nanofluidic channels have also been studied through MD simulations. A significant reduction in the electroosmotic flow due to the presence of grafted PEs was reported by Tessier and Slater.¹⁴⁵ Adiga and Brenner reported MD results for controlling electroosmotic flows as a function of effective solvent quality.¹⁴⁶

One of the major related open questions is the molecular scale understanding of the nanochannels grafted with reactive PE brushes. This “reactivity” of the PE brushes arises from their charge density being a function of the solvent pH. The relevant MD model should account for the explicit hydrogen ion concentration that has been shown, through a SCFT model,⁶⁵ to affect the PE brush behaviour. Once established, these MD models will be able to provide unprecedented understanding about issues such as fluid and ion transport, brush–brush interactions, entanglement of brushes, *etc.* in a nanochannel grafted with reactive PE brushes.

6 Techniques for fabricating a PE grafted system

6.1 General approaches for the synthesis of a brush

A polymer brush can be attached to the substrate by either physisorption or covalent-bonding-based chemisorption. Physisorption is typically observed in block copolymers (BCPs), and is possible only if one of the blocks interacts strongly with the substrate and the other block interacts weakly. In such an event, the strongly interacting block of the BCP molecule adsorbs onto the substrate. Physisorption based grafting was first reported by Hadzioannou *et al.* who demonstrated grafting of poly(styrene)/poly(2-vinylpyridine) (PS/P2VP) BCP on a mica substrate.¹⁴⁷ The P2VP exhibits preferential adsorption to mica, while PS does not adsorb. However, grafting through physisorption occurs only under very specific conditions, with a mandatory requirement being the presence of at least two distinct segments.

Additionally, the attachment in physisorption is based only on van der Waals and hydrogen bonding, which makes the process reversible. Consequently, such brushes often exhibit weak thermal and solvent stability as a result of low grafting strength.

The limitations in physisorption based grafting can be largely overcome by synthesizing the brush in such a way that there is covalent bonding between the surface and polymer molecules, which makes the grafting much more robust and stable against solvents. Covalent attachment is achieved following two distinct approaches, which are referred to as the “grafting to” and “grafting from” strategies. In the grafting to strategy, the polymer chain is first synthesized in solution with an active end group that can bind to a substrate. The reaction of such end-functional polymers with a polymer backbone precursor, containing complimentary functionality on each monomer unit, forms covalently bonded layers. The functional polymer chain can tether to the substrate with reactive groups by either coupling, radical or condensation reaction, or by physiochemical adsorption. However, due to steric repulsion between the grafted polymer chains and the incoming molecules, this method does not allow the growth of dense polymer brushes. This is one of the major drawbacks of the method.^{148,149} The existing grafted chains prevent access of the of new polymer chains to the grafting sites on the surface. The other drawback is that with the increasing polymer molecular weight, the reaction between the end-group and the complementary group of the polymer becomes less efficient. Consequently, the grafting from method is more popular, since in this approach the polymerization is initiated directly from the initiator-functionalized surface.^{150–153} The grafting to and grafting from strategies are schematically illustrated in Fig. 9, and are prepared by end-grafting of chains to or from flat or curved surfaces that are either organic or inorganic in nature. The thickness of anchored polymer brushes depends on the degree of polymerization of the polymer and the grafting density.¹⁵⁴

Apart from graft to and graft from methods of synthesizing polymer brushes on surfaces, a graft through polymerization method has also been developed. In this method, the surface is functionalized with a reacting group during the solution polymerization process. High grafting density of the polymer brush can be obtained by this simple process, under suitable reaction conditions.^{155,156}

6.2 Different polymerization approaches

Polymer brushes, grown by the grafting from strategy, allow precise control over the brush thickness, grafting density, composition, and architecture. In this method, the initiator is first anchored to the surface; followed by growth of the polymer from the initiator group. The macroinitiator has a predetermined number of initiation sites. Growth of a polymer is accomplished by exposing it to solutions containing monomers and catalysts. This method produces polymer brushes with high grafting density and greater thickness. The only constraint of the grafting from technique is that it first requires the immobilization of an initiator on the substrate. Some of the grafting from methods are anionic polymerization,^{157–159} cationic polymerization,^{160–162} ring-opening

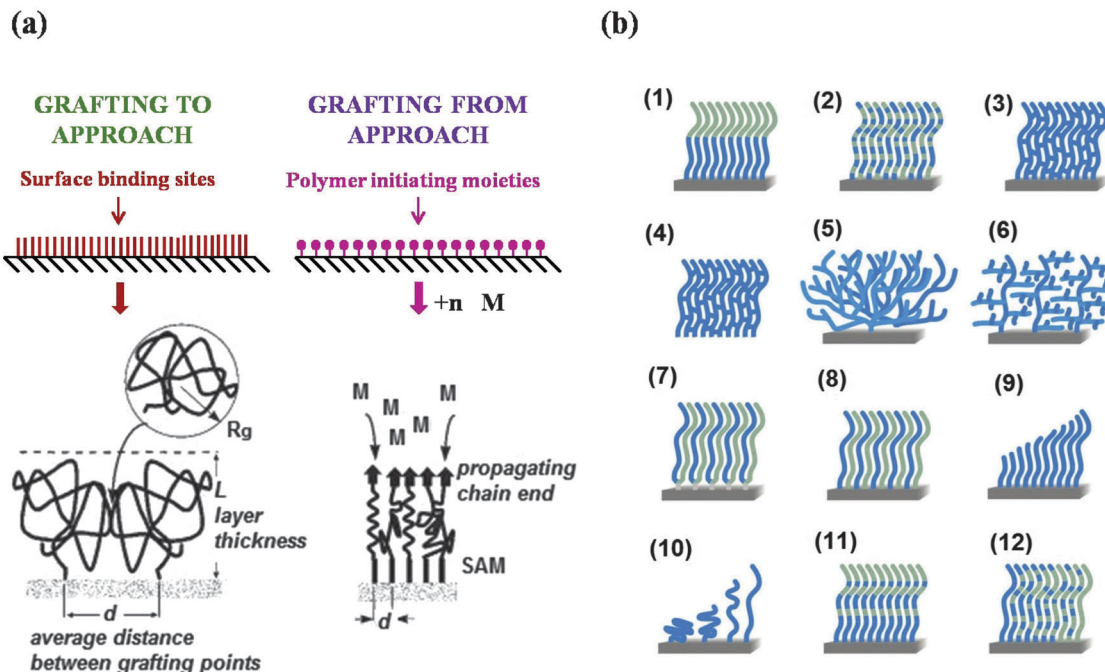


Fig. 9 (a) Pictorial representation of the grafting to and grafting from techniques with R_g as the radius of gyration of a random coiled polymer in a good solvent, L as the layer thickness, d the average lateral distance between grafting points, M an arbitrary monomer, and I an initiation site able to induce polymerization. Figure (a) has been adapted with permission from R. Jordan, A. Ulman, J. F. Kang, M. H. Rafailovich and J. Sokolov, Surface-Initiated Anionic Polymerization of Styrene by Means of Self-Assembled Monolayers, *J. Am. Chem. Soc.* 1999, **121**, 1016–1022. Copyright (1999), American Chemical Society. (b) Schematic representation of different type of brush architectures, (1) block copolymer brushes; (2) random copolymer brushes; (3) cross-linked polymer brushes; (4) free standing polymer brushes; (5) hyperbranched polymer brushes; (6) highly branched polymer brushes; (7) Y-shaped binary mixed polymer brushes; (8) standard binary mixed brushes; (9) molecular weight gradient polymer brushes; (10) grafting density gradient polymer brushes; (11 and 12) chemical composition gradient polymer brushes. Figure (b) has been adapted with permission from R. Barbey, L. Lavanant, D. Paripovic, N. Schüwer, C. Sugnaux, S. Tugulu and H-A Klok, Polymer Brushes via Surface-Initiated Controlled Radical Polymerization: Synthesis, Characterization, Properties, and Applications, *Chem. Rev.* 2009, **109**, 5437–5527. Copyright (2009), American Chemical Society.

polymerization,^{163–170} ring-opening metathesis polymerization,^{171–175} and the radical-based polymerization techniques. By ring opening metathesis polymerization (ROMP), well defined polymer brushes can be grown on a surface with appreciable thickness in a very short time. Conventional free radical polymerization has found widespread use for the synthesis of polymer brushes, but it is not possible to obtain multiblocks following this approach.^{176–188}

The most common and widely used ‘grafting from’ method for preparing polymer brushes is the surface initiated polymerization (SIP) method, which uses surface tethered initiators. By this method, the grafting density can be controlled by the proper choice of initiator and polymerization conditions. It can be applied to both flat and curved surfaces and there are wide ranges of available monomers for this method. Anionic polymerization yields good control over the thickness and formation of copolymer brushes, but the reaction conditions required are very harsh. Cationic polymerization requires a complex experimental procedure to form surface bound initiators. The reactions involved are sensitive and time consuming. On the other hand, radical polymerization proves to be an efficient method for growth of polymer brushes.^{189–191} Radical-based polymerization reactions are compatible with both aqueous and organic media and have high tolerance towards

a wide range of functional groups. A variety of methods have been developed by using radical polymerization techniques to graft polymer chains to surfaces.^{192,193} The main radical polymerization methods are atom transfer radical polymerization (ATRP), nitroxide-mediated polymerization (NMP), and reversible-addition-fragmentation chain transfer polymerization (RAFT). Atom Transfer Radical Polymerisation (ATRP) can be precisely controlled with ease of initiator formation and mild polymerization conditions. By varying the polymer brush composition and degree of polymerization surface properties can be altered significantly,^{194–196} nanopatterns can be modified^{197,198} and stimuli-responsive materials can be easily developed.^{199–201} However, the main problem in ATRP from flat surfaces is that, after halogen atom transfer to the transition metal catalyst, the concentration of the deactivator may become too low to reversibly trap the propagating radicals. This results in uncontrolled chain growth. An effective solution to this problem is by adding a persistent radical (deactivator).^{202–204} One major drawback of the ATRP process is that there is no universal recipe and hence, factors such as the choice of solvents, ligands, monomers and reaction temperature, affects the polymerization process.

Nitroxide-mediated polymerization (NMP) reaction is a useful technique for preparing styrene based polymers on surfaces. A metal based catalyst is not required in this method. NMP was

first reported by Hawker *et al.* in 1999.¹⁹⁵ They immobilized an alkoxamine based initiator on the surface of silicon to grow styrene polymers from it. The main advantage of the method is that it allows accurate control over the polymer brush molecular weight and brush thickness. However, the major drawback of NMP is the requirement of a relatively high reaction temperature of around 110 °C. Such a high temperature is not suited for certain sensitive monomers.

Reversible-addition–fragmentation chain transfer polymerization (RAFT) is another popular method of grafting polymer brushes. The main advantages of the method are: lack of catalysts, versatility of monomer choice and mild reaction conditions. This method was first reported in 2001, and since then it has been rigorously used for grafting polymer brushes using multiple approaches.²⁰⁵ The radical polymerization methods ATRP, NMP and RAFT are explained schematically in Fig. 10.

6.3 Synthesis of polyelectrolyte brushes

Although theoretically PE brushes were analyzed long back,²⁰⁶ synthesis of these brushes was not successfully demonstrated due to the lack of a universal synthesis technique like regular polymer brushes.²⁰⁷ In addition to “grafting to” and “grafting from” approaches, PE brushes can also be formed by: the adsorption of

charged block copolymers from solution, anchoring amphiphilic charged block copolymers at the air–water interface, followed by compression, and subsequent transfer of the layer to a solid substrate,²⁰⁸ or chemical attachment of charged polymers to the substrate. In the latter method, covalent chemical bonds are formed between the charged polymer molecules and the solid substrate. Although most of the work reported so far on “grafting to” technique has been on the formation “neutral” polymer brushes, there are cases where PEs have been synthesized by this method. Mir *et al.* synthesized polystyrene sulfonate (PSS) brushes terminated with trichlorosilane anchor groups on both planar^{209–211} and spherical²¹² surfaces by this technique. At first, a monolayer of neutral polystyrene molecules was formed by a reaction of the silane anchor groups with silanol groups on the substrates. This was then transformed into the PE brush *via* a polymer-analogous sulfonation reaction. PE brushes prepared by the “grafting to” technique require a two-step synthesis process as compared to neutral brush system. Therefore, there might be additional complications if the post-synthetic modification is not quantitative or side reactions might occur. A serious limitation of this process is that only relatively simple polymer molecules such as polystyrene can be surface-attached by this method. Another limitation is on the thicknesses of the surface-attached monolayers obtained by this method.^{213,214}

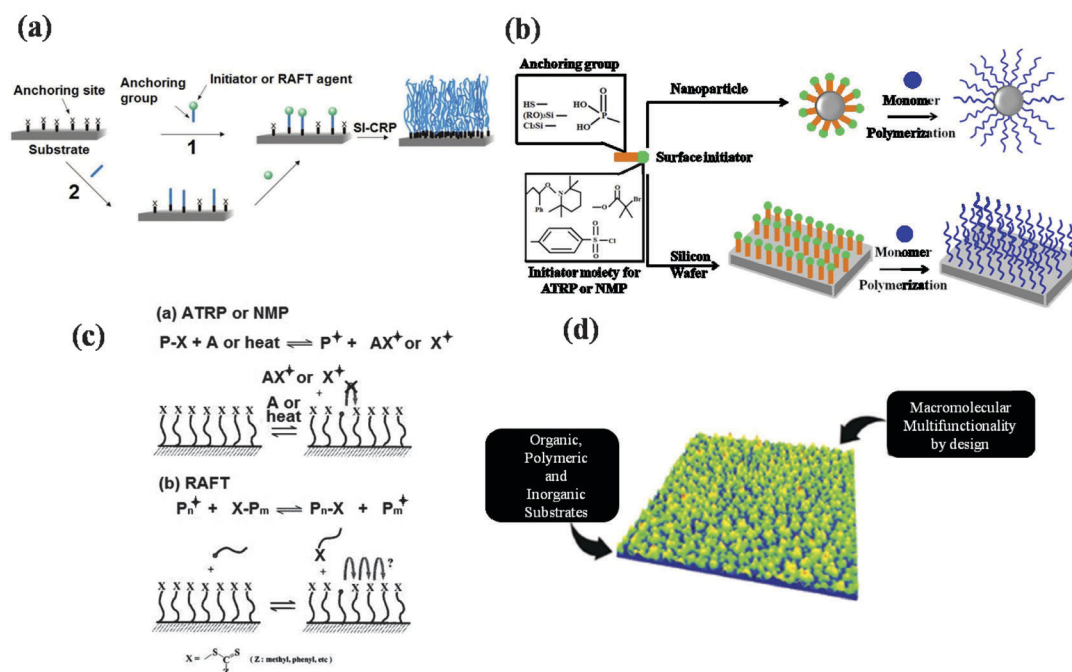


Fig. 10 (a) Brief overview of the RAFT polymerization technique for modification of the surface: (1) one step strategy; (2) two step strategy. Figure (a) has been adapted with permission from R. Barbey, L. Lavanant, D. Paripovic, N. Schüwer, C. Sugnaux, S. Tugulu and H.-A. Klok, Polymer Brushes via Surface-Initiated Controlled Radical Polymerization: Synthesis, Characterization, Properties, and Applications, *Chem. Rev.* 2009, **109**, 5437–5527. Copyright (2009), American Chemical Society. (b) Schematic representation of ATRP and NMP polymerization on the surface of metal oxide nanoparticles and Silicon wafer surfaces. (c) Comparison between ATRP or NMP, and RAFT polymerization techniques. Figure (c) has been reproduced with permission from Y. Tsujii, M. Ejaz, K. Sato, A. Goto, and T. Fukuda, Mechanism and Kinetics of RAFT-Mediated Graft Polymerization of Styrene on a Solid Surface. 1. Experimental Evidence of Surface Radical Migration, *Macromolecules* 2001, **34**, 8872–8878. Copyright (2001), American Chemical Society. (d) Atomic force microscope image (500 × 500 nm²) of poly(2-(methacryloyloxy)-ethyl-trimethyl-ammonium chloride) brushes grown on silicon substrates via surface initiated atom transfer radical polymerization. Figure (d) has been reproduced with permission from O. Azzaroni, Polymer Brushes Here, There, and Everywhere: Recent Advances in Their Practical Applications and Emerging Opportunities in Multiple Research Fields, *J. Polym. Sci., Part A: Polym. Chem.*, 2012, **50**, 3225–3258. Copyright (2012) John Wiley and Sons.

Alternatively, PE brushes can be synthesized by the “grafting from” technique, in which polymer molecules are formed directly at the surface by using a monolayer of surface attached initiators. R  he and coworkers reported the synthesis of neutral^{180–182} as well as charged polymer brushes,^{184,215} by the use of an azo initiator to start the free radical polymerization of different monomers. For the synthesis of densely grafted positively charged quaternized poly-4-vinylpyridine brushes, firstly a neutral poly-4-vinylpyridine monolayer was prepared followed by subsequent introduction of charges by a second, polymer-analogous quaternization step. The synthesized PE was found to have a high molecular mass and a high grafting density. One step syntheses of PEs have also been reported by use of the same immobilized azo-initiator monolayer. Here, the monolayers of polystyrenesulfonate or methacrylic acid were generated at the surface *in situ*.²¹⁶ This method can be extended for the synthesis of copolymer brushes; the only critical requirement being that the two monomers need to be mixed in an appropriate ratio prior to the surface-initiated polymerization step.

Ballauff and co-workers synthesized particles with a polystyrene core and a shell of polystyrene sulfonate or polyacrylic acid chains by a photo-emulsion polymerization process.^{217,218} They covered the polystyrene core lattices with a polymeric layer containing photo-initiator side-groups. PE brushes were generated on the surface of these particles by photoinitiation with the use of surface-bound initiators in the presence of a monomer. For carrying out this reaction, the polystyrene particles were dispersed in the monomer and were then irradiated in the reaction mixture. Dilution was maintained at such a level that the inter-particle distance was sufficient to prevent the recombination of radicals growing on two different particles. The brush height was reported to increase with an increasing monomer concentration during the polymerization process. The photo-emulsion polymerization technique is a convenient method for the preparation of stable PE brush modified particles. Such systems that consist of a solid core (having a diameter in the range of 100 nm) onto which long chains of anionic or cationic PEs are densely grafted are known as spherical PE brushes.

If the polymer layers become swollen when exposed to a certain environment, undesired mechanical stress may be generated. This can cause the film to detach completely or partially from the surface. Also, the polymer molecules can get displaced by other competing adsorbents from the environment under certain cases. The presence of impurities, especially multivalent ions, can alter the properties of the polymer layer. Therefore, the stability of these polymer films is of utmost importance in the practical scenario and knowledge about the swelling and interaction of the PE layer with the environment needs to be established, which is discussed next. Also, the knowledge about the swelling of PEs in humid air is required, since most of the characterization methods for determining the thickness of the polymer layer or the mass of the attached polymer are done under ambient conditions.

6.4 Characterization of PE brushes

Swelling of PE dry brushes in a humid environment. In a humid environment, the strong interaction of the PEs with water results in swelling of the polymer film. The extent of swelling is directly related to the structure of the polymer. Biesalski *et al.* reported the swelling characteristics of the poly-*N*-methyl-(4-vinylpyridinium)iodide brush in a humid environment.²¹⁵ In a typical experiment, the polymer brush was exposed to different controlled environments using a cell which was held at constant temperature and relative humidity. An increase in the film thickness was observed when the relative humidity was changed from 0 to 100 at ambient temperature. From this strong influence of humidity of the environment on film thickness we conclude that it is important to take into consideration the air humidity-induced swelling of PE layers while determining the film thicknesses. Neglecting this air humidity-induced swelling of PE layers will lead to an overestimated mass of the polymer in the film.

Swelling behavior of “strong” PE brushes. Swelling behavior of PE brushes in contact with water or salt solutions was first reported by Auroy and coworkers in 1995.²¹² In a typical experiment, the “grafting to” technique was used for the preparation of polystyrene sulfonate layers by grafting polystyrene molecules to the silica gel surface by means of chlorosilyl end-groups. The polystyrene sulfonate layers were charged by a polymer-analogous sulfonation reaction. It was observed that upon addition of small quantities of electrolyte to the solution, the brush height shrinks due to the electrostatic screening of the repulsive forces inside the layer. From mean-field theory it was predicted that the average swollen height of the PE brushes scales linearly with the length of the attached polymer.²¹⁰ Johannsmann and coworkers studied the swelling behavior of neutral poly-4-vinylpyridine (PVP) and positively charged poly-*N*-methyl-(4-vinylpyridinium)iodide (MePVP) brushes in a “good solvent” by means of multi-angle ellipsometry.^{219,220} It was observed that the swollen brush height of the neutral PVP brushes in pure water increases with graft density, whereas the height of the charged PE MePVP was found to be almost independent of the graft density. The influence of electrolytes on the internal structure of polystyrene sulfonate brush was studied by means of noise analysis of a scanning force microscope (SFM) cantilever.²²¹ It was observed that the PE brush shrinks strongly upon addition of electrolytes.

Ballauff and coworkers studied the swelling behavior of a spherical brush, which is considerably different from that of a planar surface. The swelling behavior of the spherical brush system was reported with respect to the ionic strength and the pH of the liquid phase.²¹⁸ They determined the height of the brush layer by dynamic light-scattering experiments which were in accordance with the mean-field theory. It was reported that the height of the PE brush was independent of pH at a given constant electrolyte concentration. The height of spherical brush as a function of the concentration of added monovalent salts, investigated at constant solution pH, revealed that at low ionic strength the brush height was almost independent of the external salt concentration.

However, at higher salt concentration there was shrinkage in the height of the PE brush.

Swelling behavior of “weak” PE brushes. The swelling behavior of weak PEs is different from strong PEs since in weak PEs the local charge density is not fixed but is a function of the local concentration of protons inside the brush. In fact, this specific property makes weak PE brushes most sought after for responsive surfaces. If the pH is raised or lowered new charges are generated or get disappeared due to charge recombination. The degree of dissociation of weak PEs depends on the polymer concentration, and the charge density varies within the brush as a function of the distance from the substrate. Johannsmann and coworkers investigated the polymer segment density profiles of weak polyacid brushes as a function of the pH of the environment by means of multiple-angle ellipsometry.²²⁰ It was observed that the behavior of these brushes consisting of surface-attached weak polyacid chains was significantly different from strong PEs, as the number of charges on the polymer chain varies strongly with changing pH. At very low pH the brush carries no net charge since all the sites are protonated. With an increase in pH more charges are generated inside the brush, which results in an increase in electrostatic repulsion. This causes the polymer chains to stretch out further into solution and the overall brush height increases.

Effect of pH and ionic strength on brush height of weak spherical brushes has been studied extensively by Ballauff and coworkers by dynamic light scattering experiments.^{218,222,223} They observed that at high salt concentration, the brush height shrinks in a manner similar to that of strong PEs. However, at low ionic strength the height of brush was found to increase with increasing salt concentration for a PMAA brush. Maximum brush height was observed at an intermediate concentration. For analyzing these results, the screening of charges as well as dissociation equilibrium needs to be considered. This behavior of weak PEs is somewhat counterintuitive and is sometimes referred to as the “anomalous salt effect”.

6.5 Synthesis of bimodal/mixed (PE) brushes

A bimodal brush is a homopolymer brush attached to a surface which has two different lengths. When the chains are of two different materials, they are known as the mixed bimodal brush. If one or both of the polymers in the mixed brush carry electrical charge, much more complicated behaviour is observed.²²⁴ In equally charged homopolymer brushes, the intra-chain and inter-chain Coulombic repulsive interactions lead to stretching of the chains. On the other hand, in oppositely charged mixed brushes the screening of electrostatic repulsion depends on the degree of charge compensation of the total charge on homopolymer A by the oppositely charged homopolymer B. When the total charge of A is greater than B, polymer chains of A are coiled due to the electrostatic attraction between A and B. When the total charge of A is equal to that of B, both chains form a compact brush. For synthesis of mixed brushes, two chemically distinct homopolymers are tethered to a surface such that there is selective solvation of one of the chain types. Both “grafting from” and “grafting to” methods can be used

for the synthesis of mixed brushes. The synthesis of mixed PEs is a two-step process: in the first step, neutral homopolymer brushes are synthesized by the “grafting to” or the “grafting from” method. In the next step, the second homopolymer is grafted followed by preparation of the mixed PE brush from the mixed neutral brush by a polymer analogous reaction.

Synthesis of mixed PEs by the “grafting to” method was employed by Minko and coworkers for the synthesis of mixed polystyrene (PS)–poly(2-vinylpyridine) (P2VP) brushes by a two-step process.²²⁵ In the first step, carboxyl terminated PS was grafted onto the surface of silicon wafer functionalized with (3-glycidoxypentyl)trimethoxysilane (GPS). This was followed by the grafting of carboxyl terminated P2VP onto this surface. In the next step the mixed PE was obtained by protonation of P2VP in an acidic aqueous solution. Polyacrylic acid (PAA)–P2VP mixed brushes were prepared by a similar method, by grafting of carboxyl-terminated poly(*tert*-butyl acrylate) (PtBuA) and P2VP, followed by the hydrolysis of PtBuA in the presence of *p*-toluene sulfonic acid.

Similarly, Minko and coworkers reported the synthesis of mixed PEs by the “grafting from” method.^{226,227} A typical experiment involved the functionalization of the surface of Si wafer with GPS followed by treatment with diaminopropane. The initiator for the reaction, an acid chloride of 4,40-azobis-(4-cyanopentanoic acid) (Cl-ABCPA) was then chemically grafted to this modified Si wafer surface. The monomers, styrene and 2-vinylpyridine, were then polymerized onto the surface followed by the synthesis of the mixed PEs in a manner similar to that employed in the grafting to technique.

6.6 Multilayer PE brushes

Another fascinating field of polymer brushes is the development of multilayers of PEs on surfaces. The multilayer of PEs shows promising application in the field of chemistry, materials science and biology. PE multilayers were first developed by a layer-by-layer deposition technique proposed by Decher *et al.* in 1991.²²⁸ This layer-by-layer technique is a simple and powerful method for fabricating multilayer brushes. For the formation of multilayer brushes, the substrate is alternatively dipped into two different PE solutions. One of the solutions consists of a positively charged PE and the other of a negatively charged PE. In each of these dipping steps a new PE monolayer was adsorbed. Electrostatic interactions govern the formation of these multilayers and hence the driving force for each monolayer deposition is charge overcompensation. The thickness of these layers depends on the surface charge density, charge density of the adsorbed PE layer and the ionic properties of the PE solution.^{229,230} Thickness is independent of the number of already deposited layers. It has been found to be of the order of 0.5 nm.²³¹ This small increase in the thickness with each deposition step is advantageous when there is a requirement for thin multilayer assemblies, but is quite slow when thick multilayers are desired, as it becomes a very time consuming process. The other concern is the stability of the entire PE multilayer assembly, since it uses a charged surface as a substrate. This is also not a very robust technique since the

attachment of the first layer depends totally on the interaction of the polymer with the substrate surface charge.

6.7 Patterned polymer brushes

Another interesting class of materials that are excellent candidates for studying the responsive properties of the PEs are the nano patterned polymer brushes, which are fabricated by combining the essential concepts of soft lithography and grafting from brush synthesis approach. Typically a patterned initiator surface is first produced. Micro contact printing or dip pen nanolithography is a suitable technique for fabricating a patterned SAM layer, which can be subsequently amplified by ATRP or other polymerization techniques. For example, Husemann *et al.* first micro contact printed a gold surface with a non reactive thiol ($\text{CH}_3-(\text{CH}_2)_{15}\text{SH}$) following standard micro contact printing. This chemically patterned surface was subsequently immersed in a solution of a functionalized thiol ($\text{HO}(\text{CH}_2\text{CH}_2\text{O})_2-(\text{CH}_2)_{11}\text{SH}$). This resulted in the open spaces on the surface left after the first patterning step getting filled up by the second thiol. In the final step, surface-initiated ring opening polymerization (ROP) of ϵ -caprolactone was performed over the functionalized SAM domains to result in a patterned polymer brush.¹⁶⁴ Patterned polymer brushes are also key to the fabrication of a binary polymer brush, either by sequential polymerization over either the SAM coated and uncoated areas or over using a dissolvable polymer mask itself. For example, Prucker and co workers created surface patterns by selective UV irradiation of homogeneous SAM layers of a photo sensitive azo initiator, which allowed repeated surface initiated polymerization (SIP), leading to chemically heterogeneous polymer-brush surfaces.^{183,232,233} Zhou *et al.* on the other hand reported the possible fabrication of a chemically tethered binary polymer-brush pattern based on a two-step surface-initiated ATRP technique, with an intermediate UV induced selective etching of the first brush layer, exposing the bare substrate. This method is extremely generic and can be applied for fabricating a variety of binary brushes such as PHEMA (poly hydroxyethyl methacrylate)/PMMA (poly methyl methacrylate), PHEMA/PDMAEMA (poly dimethylaminoethyl methacrylate).²³⁴ AFM morphologies of PHEMA/PMMA and PHEMA/PDMAEMA binary polymer brushes and the corresponding cross-section profiles are shown in Fig. 11. Similar alternating structures with PE brushes comprising of positively charged poly(methacryloyloxy)ethyl-trimethylammonium chloride (PMETAC) having quaternary ammonium groups and negatively charged polymethacryloyl ethyl phosphate (PMEP) bearing phosphate groups were synthesized following a two stage ATRP process, with a micro contact printed initiator layer for the first reaction. Such a surface could be used as a seeding template for subsequent site-selective electrodeless deposition of Cu/Ni bimetallic patterns, based on the ability of the individual domains to bind to the oppositely charged metallic complex *via* an ion-pairing interaction.²³⁵

6.8 Approaches for grafting nanoparticles

Polymers grafted on nanoparticles significantly influence the particle behaviour and spatial distribution. Polymers can be

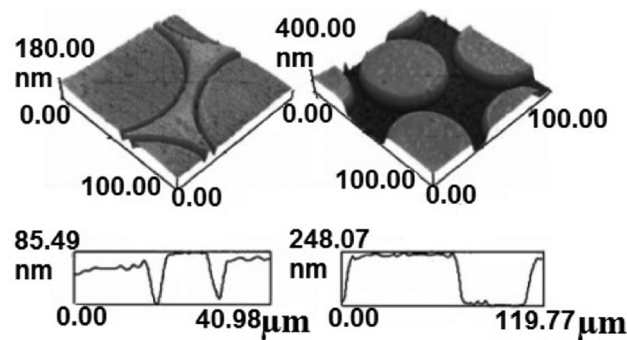


Fig. 11 AFM morphologies of binary polymer brush patterns of 90 nm PHEMA/60 nm PMMA (the left figure) and 45 nm PHEMA/260 nm PDMAEMA (the right figure) and their cross-section profiles. Both parts of the figure have been reproduced with permission from F. Zhou, L. Jiang, W. Liu and Q. Xue, Fabrication of Chemically Tethered Binary Polymer-Brush Pattern through Two-Step Surface-Initiated Atomic-Transfer Radical Polymerization, *Macromol. Rapid Commun.*, 2004, **25**, 1979–1983. Copyright (2004) John Wiley and Sons.

coated onto nanoparticles by different methods. Polymer molecules can be physisorbed onto nanoparticles for preventing agglomeration and maintaining a stable dispersion.²³⁶ Coating of PEs on nanoparticles is another extensively studied area. Here PEs are coated on nanoparticle surfaces of opposite charge. There exists coulombic force of attraction between the particle and the PE layer.^{237–240} Multilayers can be coated on the surface by using PEs of opposite charge of the deposited layer. Thus the thickness of the PE coating can be easily controlled. Most commonly, polymers are attached to nanoparticles through covalent bonds. Since these covalent bonds are stronger in nature, the performance of the grafted nanoparticles is better. This falls as usual under graft to and graft from approaches. The grafting to method is quite simple and modular since the step involved is attachment of an end functionalized polymer to the surface.^{241,242} However, there is a problem of strong steric repulsion between the attached chain and the diffusing chain. This factor hinders the grafting density. In the other method, that is the grafting from method, the polymer is grown from the surface of the particle with the help of an initiator. The problem of steric repulsion is eliminated in this method. The different polymerization methods employed are: cationic,^{236,243,244} anionic,^{237,238} ring-opening metathesis,²³⁹ and controlled radical polymerization methods.^{240,245–250} Of these methods the widely used method for grafting nanoparticles is the controlled radical polymerization technique. These techniques include, ATRP, NMP and RAFT as discussed previously.^{192,193} With these techniques, precise control over the chain composition, molecular weight, architecture and chain functionality is possible. Reversible addition–fragmentation chain transfer polymerization (RAFT) has an advantage over the other methods, that is, it is free from contamination of the metal catalyst.

Grafting of responsive brushes on nanoparticles, thereby making them responsive to an external stimuli, is another important area of research. Armes and co-workers grafted thermoresponsive PNIPAM brushes on silica particles by the

ATRP technique.²⁵¹ These PNIPAM brushes were also grown from polystyrene latex particles by Kizhakkedathu *et al.* via surface-initiated ATRP.²⁵² Poly(*N,N*-dimethylaminoethyl methacrylate) brushes that respond to both pH and temperature were grafted on colloidal particles by Zhang and co-workers.²⁵³ Recently, transport of particles grafted with polymers across two different phases has been possible. This transport of nanoparticles utilizes the temperature induced transition in the brushes from the hydrated to dehydrated state.²⁵⁴ Horton and co-workers synthesized temperature and pH sensitive particles to study their reversible phase transfer behaviour between water and a hydrophobic ionic liquid. Such systems are very useful in drug delivery across amphiphilic biological membranes.²⁵⁵ A responsive system based on mixed polymer brushes was demonstrated by Minko and co-workers. The mixed copolymer brush constituted of P2VP-*b*-PEO and P2VP-*b*-PS units, which showed a drastic response to external stimuli.²⁵⁶

Another important area of interest is the distribution of the number of chains per particle. Various models have been developed to quantify the distribution of grafted polymers per particle. Characterizing the variation in the number of chains grafted from particle to particle has proven to be a challenging task. Some significant results have been reported by Bockstaller and co-workers and Vaia and co-workers in this field.^{257–259} It has been shown that grafting density variation from particle to particle is a function of particle radius and significant polydispersity effects exist upon decreasing the particle radius. One area which still remains unexplored is the spatial distribution of the graft points on the particle surface. Research in this area is equally important for controlling the grafted chain structures.

6.9 Approaches for grafting in nanochannels

Nanochannels are capable of controlling and manipulating transport of ionic species through them.^{260–265} Nanochannels with significant electrostatic interaction show transport properties due to surface charges. Such properties include concentration enhancement and depletion and surface charge governed transport.^{266–271} Although, there are varied fascinating uses of nanochannels in microfluidics which are difficult to fabricate and hence have attracted attention from chemists, physicists and engineers.^{93,272–274}

Apart from the development of conventional nanochannels, nanochannels with supramolecular systems which are sensitive to external stimuli have captured the attention of scientists. These nanochannels with “smart” transport properties respond to external stimuli and modulate the surface charges.¹⁰⁷ Surface-initiated controlled radical polymerization techniques are mostly used to synthesize polymer brushes in nanochannels. Surface initiated atom transfer radical polymerization is the most favourable one because of its major advantages such as, reproducibility, robustness of the chains, controlled grafting density and well-defined chain length and architecture.^{275–278}

Flow manipulation in nano and microchannels by functionalizing the surface has been studied extensively over the past decade.^{279–282} Nanochannels in which pore size and permeability

can be manipulated in response to external stimuli are a new class of smart materials that have potential application in drug delivery and flow control in microfluidic chips, where fluid transport can be regulated on demand. They have been successfully employed to switch interfacial energy in response to external stimuli such as temperature,²⁸³ pH,^{179,282,284} ionic or solute concentration,^{178,285} and light.^{286,287} The three distinct approaches for functionalizing nanochannels with smart polymers are: polymer chains are end grafted onto the surface to form a brush layer inside the nanochannel. This layer stretches and collapses in a reversible manner in response to external stimuli, for opening and closing the pores. End grafted polymer brushes exhibit superior response time since they respond very efficiently to external stimuli.²⁸⁸ In the second method, responsive polymer gels are incorporated into a nanoporous support. These polymer gels have a crosslinked configuration. In the third method, PEs are deposited in a layer-by-layer assembly inside the nanochannel. These PE layers respond to external stimuli by changing their film thickness; and hence this property is utilized to produce membrane responsiveness.

7 Applications

7.1 Responsive brushes

For a typical gradient brush surface, composition and properties change continuously along the surface coordinate. Adaptive and responsive surfaces have recently attracted the attention from researchers because of their prospective applications in the field of micro- and nanofluidics, transport of liquids, controlled drug release, nano and bio-tribology, controlled cell growth and proliferation, bio- and chemosensing, and bioelectronics.^{289–295} The gradient surfaces possess a directional continuous variation of any of these following properties: number of functional groups,²⁹⁶ molecular weight,²⁹⁷ chemical composition,²⁹⁸ grafting density,^{154,299} or nanoparticle concentration.³⁰⁰

Planar gradient-responsive surfaces. Gradient brushes can be created by either creating a gradient of the initiator on the surface, or by gradually immersing the sample having a uniform layer of the initiator in the monomer solution. One can synthesize brushes where the responsive behaviour is inherent of the polymer. The response is hence uniform for all the polymer chains. Examples include PEs and thermoresponsive polymers. They respond to different stimuli such as temperature, nature of the solvent, pH, ionic concentration, electric field and magnetic field. Strong PEs are insensitive to local pH and added salt has no effect unless the ionic strength of the solution approaches the level of the ionic strength inside the brush. Thus, strong PEs are useful for design of responsiveness to humid and aqueous environments when the high degree of swelling of the brush in water or a humid atmosphere results from a strong osmotic pressure of trapped counterions. Weak PEs demonstrate remarkable responsiveness to changes in external pH and ionic strength. Weak PEs carrying basic functionalities expand upon a decrease of pH, whereas, acidic PEs expand upon an increase of pH. At high salt concentration, weak PEs shrink in a manner similar to strong PEs. However, at

certain values of pH, the weak PEs shrink in the absence or the presence of very less concentration of salt. At low ionic strength, the electrostatic interaction results in condensation of counterions in the brush; this is known as the neutral polymer brush regime. Upon increasing the ionic strength, screening of the electrostatic interaction occurs and the system is present in the osmotic regime.²¹⁸

The other type of responsive brush is the block copolymer brush. In this, the responsive behaviour originates from phase segregation of the different blocks in response to different external stimuli.³⁰¹ Although PE block-copolymer brushes have been less investigated, it has been predicted that electrostatic interaction can incorporate additional responsive mechanisms to the block copolymer system.³⁰²

Lastly, it is possible to synthesize mixed polymer brushes, where responsive behaviour can be controlled by intricately balancing the layered and the lateral phase segregation for two or more different chains randomly grafted to the same substrate. In mixed PE brushes, electrostatic interactions are used to regulate the mechanism of the phase segregation. Weak mixed PEs form a neutral complex at the isoelectric point, while outside the isoelectric point, one of the PEs dominates in the topmost layer and controls the electrical charge of the mixed brush. Thus, the surface composition of the mixed PE brush can be switched by changing the external pH.^{224,303,304}

Responsive polymer brush coatings grafted onto porous membranes have been successfully used to gate liquid flow.^{179,282,305,306} Mixed polymer brushes that respond to external stimuli are capable of manipulating the wetting behaviour of fluids in microchannels. This stimuli responsive property of mixed polymer brushes has been exploited extensively by Stamm and co-workers.^{226,307,308} The response mechanism originates from the preferential phase segregation to the top of one of the two polymers, in the presence of an external stimuli (solvent, pH, temperature).³⁰⁹ Poly(*N*-isopropylacrylamide) (poly(NIPAAm)) is a temperature-sensitive polymer that has been successfully grafted to channels of microfluidic devices for flow control. The chains of poly(NIPAAm) remain in an extended configuration below its lower critical solution temperature of 32 °C, and acquire a reversible collapsed hydrophobic coil like structure above its LCST. Using this unique property of this polymer of responding to changes in temperature, thermally actuated valves and gates were designed by Yu *et al.* They developed porous polymer monoliths with poly(NIPAAm) chains grafted to the pore surface.²⁸³ Okano and co-workers designed microflow valves with these poly(NIPAAm) grafted chains. These valves were not designed by scaling down conventional valve designs for changing the bulk hydration structure. Hydration of these grafted chains at the capillary interfaces increases microviscosity at the wall interfaces, which influences the flow frictional resistance.³¹⁰ Tethered polymer chains at microchannel interfaces permit control over the interaction of the surface with fluids, using their reversible temperature-induced properties.³⁰⁹ Flow regulation by thermo-responsive polymers is a function of grafting density, degree of polymerization, and ionic concentration.

Similarly, pH responsive poly(4-vinyl pyridine) P4VP brushes were incorporated into nanochannels by Azzaroni and co-workers for controlling the transport properties by manipulating the proton concentration in the surrounding environment. The ionic current switching characteristics of these nanochannels resemble biological channels where pH-dependent transport is observed.³¹¹ Ito *et al.* investigated pH-triggered water permeation control through grafted membranes with the use of poly(acrylic acid), poly(methacrylic acid) and poly(ethylacrylic acid). There was a change in permeation of the membrane at pH values of 3.0, 4.0 and 6.8 for poly(acrylic acid), poly(methacrylic acid) and poly(ethylacrylic acid) respectively. They also studied water permeation behaviour through a membrane, incorporated with the poly(L-glutamic acid) brush. Poly(L-glutamic acid) brushes undergo a helix-coil transition in response to pH. At low pH values, it has a helical conformation, whereas at high pH the helical conformation expands to a random coil conformation. This transition reduces the effective pore diameter and thus affects the system permeability.¹⁷⁸ Mika *et al.* synthesized membranes with inverse pH behaviour. They showed that polypropylene microfiltration membranes grafted with poly(4-vinylpyridine) exhibit a large chemical valve effect. There was a significant increase in the pressure driven permeability when the pH was increased from two to five.³¹² Qu *et al.* designed a novel membrane system by incorporating both grafted polymers and gels. It consisted of a porous polyvinylidene fluoride (PVDF) membrane grafted with positive pH-responsive poly(methacrylic acid) (PMAA) brush layers. A negative pH-responsive hydrogel, poly(*N,N*-dimethylaminoethyl methacrylate) (PDM), functioned as a pumping element. The collective action of the brush layers and hydrogel produced much higher response rates than either of the mechanisms used alone.³¹³

7.2 Nanoparticles with responsive brushes

Incorporation of nanoparticles into the polymer brush creates a repulsive pressure, which has an effect opposite of the interaction potential between the particles and the tethered chain functional groups.³¹⁴ Thus, the key is to fine tune the interaction between the tethered chain groups and the nanoparticles, which determines whether the nanoparticles will be dissolved in the brush or will remain as insoluble moieties.

By grafting polymers onto nanoparticles, the interaction between the particles can be significantly regulated, which influences the physical properties of the nanoparticle dispersion. These interparticle interactions can be tuned by grafting responsive brushes onto nanoparticles.^{4,315–318} Spherical brushes grafted onto nanoparticles behave similar to polymer brushes on planar surfaces if the radius of the particle is greater than the brush length. However, the situation becomes more complicated when the radius of the particle is smaller than the brush thickness. The polymer brush has more free space available at the periphery as compared to the surface. As a result of this relaxed geometry of the brush, a radial concentration profile in the polymer brush is created. It has been reported in the literature that hybrid multiresponsive tethered

block copolymer systems combined different responsive mechanisms in one particle.^{217,319}

The high surface area to volume ratio of nanoparticles facilitates its application in varied fields of science and technology. Grafted polymers help in stabilizing these particles, due to the steric interactions of the polymer chains. Grafted nanoparticles have potential application in the field of sensors, optics, electronics, drug delivery, catalysis, control of fluid rheology and biomedical devices.^{320–324} Grafted nanoparticles behave in a similar fashion to surfactants and exhibit surface active properties. Surface activity can also be tuned by changing the graft density and solvent quality.³²⁵ By surface modification of nanoparticles with functional polymer brushes “smart” colloidal systems can be created. The dispersion behaviour of these smart systems can be altered by manipulating the solvent, temperature, pH, or light.^{246,326–328} Such stimuli responsive particles find application due to their precise on–off control. Stabilization of pickering emulsions with nanoparticles modified with thermo responsive polymer brushes was studied by Matyjaszewski and co-workers. It was demonstrated that PDMAEMA brushes grafted on silica nanoparticles stabilized xylene-in-water and cyclohexane-in-water emulsions with very low grafting densities.²⁹⁷ Ion responsive, cationic poly(2-(methacryloyloxy)-ethyl-trimethyl-ammonium chloride) (PMETAC) brushes grafted on silica nanoparticles were used by Huck and co-workers for stabilizing emulsions. PMETAC brushes were able to tune surface hydrophilicity because of their counterion specific properties.³²⁹ Apart from homopolymers, mixed polymer brushes grafted on nanoparticles help in stabilizing emulsions, regulating particle transport and fabrication of ultrahydrophobic coating.²⁵⁶ An optically responsive polymer, poly(spiropyran), was grafted onto silica colloids by Wu and coworkers to study their change from the hydrophobic to hydrophilic state upon irradiation with UV light.³³⁰ These colloids can be reversibly transferred from an oil phase to a water phase by shining UV or visible light on the dispersion. These colloids can act as particulate emulsifiers, since they demonstrate both hydrophilic and hydrophobic behaviour at intermediate irradiation times.

Ohno introduced a new class of grafted particles, known as semisoft systems. These particles are grafted with a concentrated layer of polymer brushes. The driving force for crystallization in these systems is the long range repulsive attraction between the brush layers.³³¹ The inter-particle distance and crystal structure of these semisoft colloidal crystals can be controlled by manipulating the graft chain length and density. These colloids can be used in anti-biofouling applications.³³²

Bioresponsive polymers grafted on nanoparticles are slowly gaining popularity because of the many drawbacks of the conventional drug delivery method. The main drawback is the lack of precision in the drug delivery system. Polymers have been effectively employed in tailoring drug release because they are capable of maintaining drug concentration within the desired therapeutic range. However, they do not respond to metabolic changes in the body and are not able to modulate drug release at the area of interest. Due to the above reasons, bio-responsive polymers are now used as drug carriers.

Responsive polymers are sensitive to external stimuli, such as temperature, light, magnetic field, and ultrasound, and help in precise release of the drug at the desired location. This is done by tuning the chemical moieties of the polymer.^{333,334}

Asymmetrically coated particles, that is, particles whose surface chemical composition differs on the two sides are known as Janus particles. These particles show a high degree of anisotropy. They are used as emulsion stabilizers, micro-rheological probes and functional elements for design of electronic paper devices. Stimuli-responsive Janus particles decorated with polymer brushes with different polarity and charge can assemble and disassemble in response to change of environmental conditions.³³⁵ These responsive Janus particles can be synthesized by sequential grafting of two types of polymers on the silica core using a combination of “grafting-from” and “grafting-to” approaches.³³⁶ These two polymers grafted on the particles can differ in terms of hydrophobicity, polarity and responsive behaviour. Janus particles comprising of two oppositely charged PEs – polyacrylic acid (PAA) and poly(2-vinylpyridine) (P2VP) – show reversible aggregation-stabilization behaviour. With changes in pH, the electrostatic repulsion and attraction between the two parts of the particle get altered. Switchable Janus particles decorated with two different responsive polymers find application in stabilization of emulsions and regulation of the molecular transport at the interface. The various applications of grafted nanoparticles are schematically shown in Fig. 12.

7.3 Surface wettability modification

Manipulation of surface properties by the use of stimuli responsive polymer brushes grafted onto surfaces is an extensively studied area. These surfaces are able to show reversible and non reversible changes in wettability with a change in external conditions, such as temperature, solvent, counterion nature and light.³³⁷

Temperature-driven changes in wettability. Temperature variation can promote significant changes in the wetting properties of thermo-responsive polymer brushes. PNIPAM has been extensively used to control the wettability of surfaces.³³⁸ Below the lower critical solution temperature, LCST, PNIPAM chains remain in a swollen configuration due to the inter-molecular hydrogen bonding between the PNIPAM grafted chains and the surrounding water molecules. This promotes hydrophilicity in the PNIPAM chains. At temperatures above the LCST, the chains remain in a compact and collapsed configuration, thus increasing the hydrophobicity. Grafting density and the molecular weight of the PNIPAM chains play an important role in these temperature driven changes.^{339–341} Apart from PNIPAM, copolymers of 2-(2-methoxyethoxy)ethyl methacrylate (MEO2MA) and oligo(ethylene glycol)methacrylate (OEGMA) have been found to be useful as thermo-responsive polymers. In these copolymers, the change from hydrophilicity to hydrophobicity is independent of molar mass, concentration and ionic strength.^{342–344} Another class of materials that show thermo-responsive behaviour are the sulfobetaines.³⁴⁵

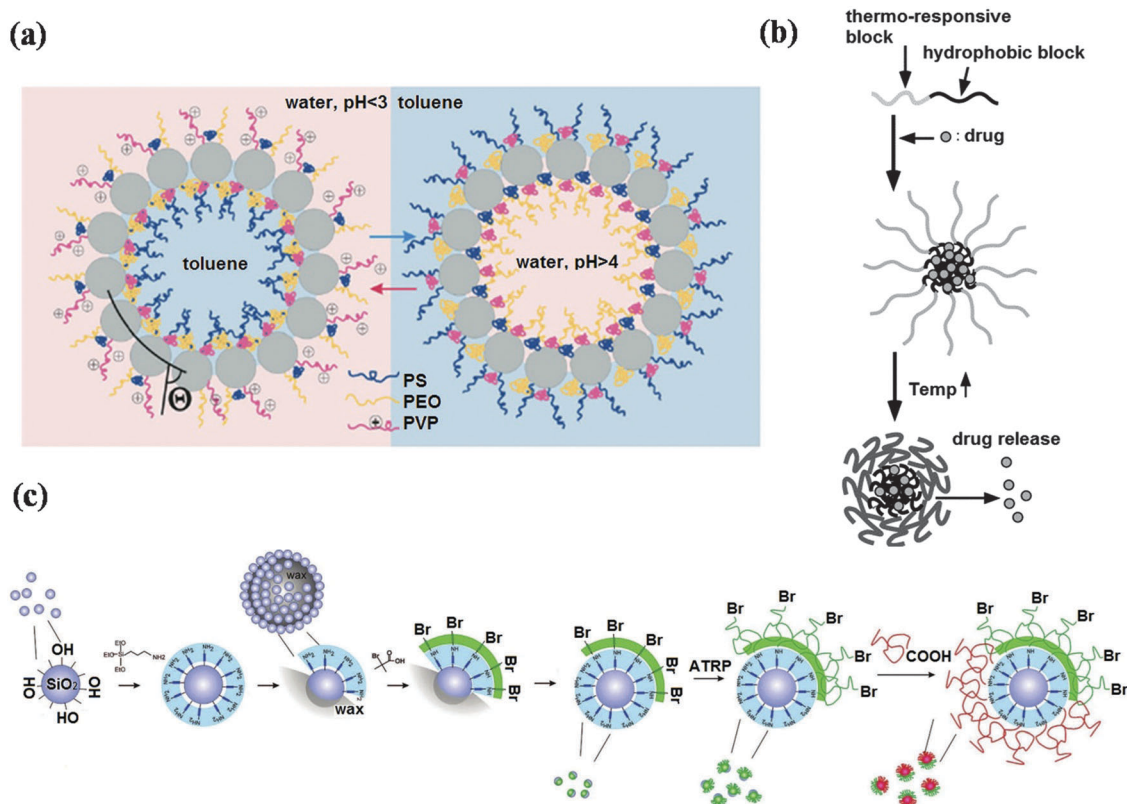


Fig. 12 (a) Oil-in-water and water-in-oil emulsion stabilized by responsive hairy nanoparticles. Figure (a) has been reproduced with permission from M. Motornov, R. Sheparovych, R. Lupitsky, E. MacWilliams, O. Hoy, I. Luzinov and S. Minko, *Adv. Funct. Mater.*, 2007, **17**, 2307–2314. Copyright (2007) John Wiley and Sons. (b) Thermoresponsive polymeric micelle for selective drug release at the appropriate temperature. Figure (b) has been reprinted from M. Nakayama, T. Okano, T. Miyazaki, F. Kohori, K. Sakai and M. Yokoyama, *Molecular design of biodegradable polymeric micelles for temperature-responsive drug release*, *J. Controlled Release*, 2006, **115**, 46–56. Copyright, with permission from Elsevier. (c) Preparation of bicomponent Janus particles by grafting from and grafting to approaches. Figure (c) has been reproduced with permission from S. Berger, A. Synytska, L. Ionov, K-J Eichhorn and M. Stamm, *Stimuli-Responsive Bicomponent Polymer Janus Particles by “Grafting from”/“Grafting to” Approaches*, *Macromolecules*, 2008, **41**, 9669–9676. Copyright (2009), American Chemical Society.

Solvent-induced changes in wettability. Surface properties of polymer brushes vary significantly from their bulk properties. This is particularly pronounced in the case of multicomponent systems, where one of the chains has preferential affinity for the interface. This in turn affects the wettability of the surface, and by simply using a solvent preferential to one of the polymers, it is possible to switch between the hydrophilic and the hydrophobic state.^{307,315,346} If we have a block copolymer assembly, the effect of solvent on the wettability is quite different, since the chains are connected to each other.¹⁹⁷ For example, if the poly(methyl methacrylate)/poly(styrene) block copolymer (PMMA-*b*-PS) is exposed to cyclohexane or dichloromethane, the PS chains preferentially come to the surface. However, the PS chains are not mobile because they are attached to the PMMA chains at one end. Thus, the PS chains form an ellipsoid configuration which increases the hydrophobicity of the surface.³⁴⁷ Brittain and co-workers studied the solvent responsive behaviour of triblock copolymers.³⁴⁸ They demonstrated that in PMA-*b*-PS-*b*-PMA, the middle block showed different physicochemical properties as compared to the end blocks. When the polymer was exposed to dichloromethane, hydrophilic PMA was closest to the surface and thus

the film showed a contact angle of 69°. When the triblock copolymer was exposed to cyclohexane, the polymer brushes underwent a collapsed configuration and showed hydrophobic nature.

Ion and pH-induced changes in wettability. In this category of tuning wettability, PEs are extremely useful. This is because the surface properties of PEs are determined by the nature of the ions to which they are exposed. Surface effects can be modified by polyelectrolytes because the conformation of these polymers is affected by electrostatic interaction, solvation and excluded volume effects. Charge repulsion, arising due to electrostatic interaction and excluded volume effects, governs the wettability of the surface in the absence of an electrolyte. This renders the surface hydrophilic. In electrolytes, electrostatic screening of the cationic charge becomes dominant and results in collapse of the chains into a coiled conformation. This renders the surface hydrophobic.³⁴⁹ Polymers containing acidic or basic groups are pH responsive in nature because a change in pH influences their morphology and surface charge. At high pH, we get fully ionized brushes, which make the surface hydrophilic. At low pH, the brushes are protonated and thus the surface is hydrophobic. At intermediate or neutral pH, the surface has intermediate hydrophobicity.³⁵⁰

Light-induced changes in wettability. This research area that uses light as stimulus for modifying the wettability of a surface gained importance because of its ability to bring about a change in each monomeric unit within a polymer film, which in turn, significantly influences the wetting properties. The use of spiropyran containing polymer brushes, a photosensitive functional group, was first reported by Locklin and co-workers.^{351,352} Under the influence of light, the geometry of the spiropyran molecule undergoes a drastic change from ring closed configuration to ring opened configuration. This change is accompanied by a change in dipole moment and consequent modification in the wettability of the surface. Apart from this, the use of polyisoprene, poly(2-vinylpyridine) and PMAA as photoswitchable polymers has been reported in the literature.^{226,353} Brown and co-workers incorporated light-activatable photoacid generators (PAGs) into block copolymer brushes. This

results in the generation of an acid, the activation of which initiates the conversion of poly(*t*-butyl methacrylate) (PtBMA) blocks into poly(methacrylic acid) (PMAA) domains. This causes a change in wettability from the hydrophobic to hydrophilic state.³⁵⁴

Surface wettability modification with mixed brushes. A surface grafted with a mixed polymer brush can adapt to the properties of either of the two polymers depending on the external stimuli.^{4,11,226} One of the polymers comprising the mixed brush is hydrophilic in nature, whereas the other polymer is hydrophobic. The chains of the hydrophilic polymer remain in a swollen state when exposed to its selective solvent, while the hydrophobic chains remain in a collapsed configuration. The result of this complex interplay causes a morphology change and the hydrophilic polymer is driven to the surface. This renders the surface hydrophilic.³⁰⁸ On the other hand, if the surface is exposed to a non-polar solvent, the situation is just the

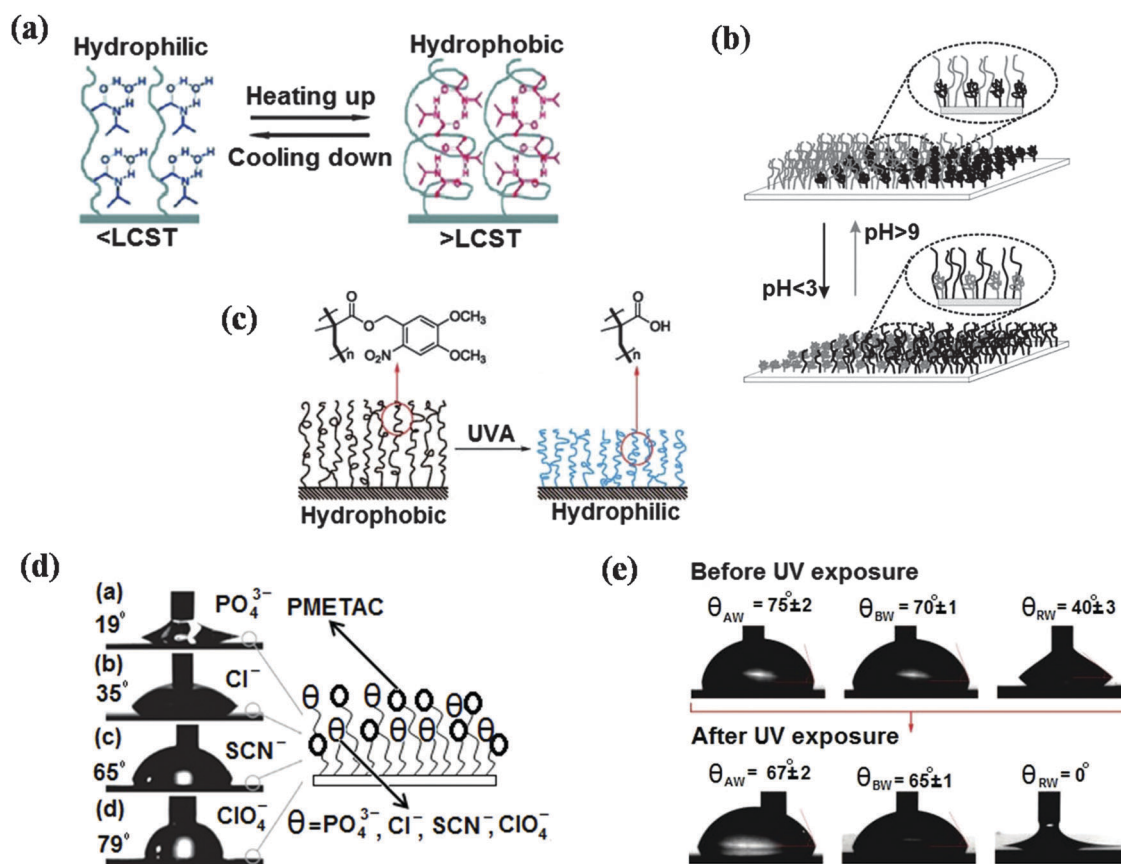


Fig. 13 (a) Schematic representation of the conformational change in the PNIPAAm brush induced by temperature. Figure (a) has been reproduced with permission from X. J. Feng and L. Jiang, Design and Creation of Superwetting/Antiwetting Surfaces, *Adv. Mater.*, 2006, **18**, 3063–3078. Copyright (2006) John Wiley and Sons. (b) Switching behaviour of mixed PE brushes upon change in pH. Figure (b) has been reproduced with permission from L. Ionov, N. Houbenov, A. Sidorenko and M. Stamm, Inverse and Reversible Switching Gradient Surfaces from Mixed Polyelectrolyte Brushes, *Langmuir*, 2004, **20**, 9916–9919. Copyright (2004), American Chemical Society. (c) UVA induced change in the PNVOCMA brush to PMAA. Figure (c) has been reproduced with permission from A. A. Brown, O. Azzaroni and W. T. S. Huck, Photoresponsive Polymer Brushes for Hydrophilic Patterning, *Langmuir*, 2009, **25**, 1744–1749. Copyright (2009), American Chemical Society. (d) Changes in wettability of an Au surface, modified with PMETAC brushes in the presence of the following counterions: (1) PO₄³⁻, (2) Cl⁻, (3) SCN⁻, (4) ClO₄⁻. This figure also includes the structure of PMETAC brushes. Figure (d) has been reproduced with permission from O. Azzaroni, A. A. Brown and W. T. S. Huck, Tunable Wettability by Clicking Counterions Into Polyelectrolyte Brushes, *Adv. Mater.*, 2007, **19**, 151–154. Copyright (2007) John Wiley and Sons. (e) Change in contact angles of the PNVOCMA brush before and after 10 min UVA exposure. Figure (e) has been reproduced with permission from A. A. Brown, O. Azzaroni and W. T. S. Huck, Photoresponsive Polymer Brushes for Hydrophilic Patterning, *Langmuir*, 2009, **25**, 1744–1749. Copyright (2009), American Chemical Society.

opposite as discussed. The hydrophobic polymer will be present at the surface, thus, rendering the surface hydrophobic. These changes are completely reversible and intermediate states are achieved in less selective solvents. Some of the necessary conditions required for switching behaviour are: the chains should be of moderate length, the chains should have a certain degree of mobility, the solvent should not be a complete non-solvent for the other component, and the grafting density should not be too high. The various schemes of stimuli induced surface wettability modification are shown in Fig. 13.

8 Future developments

While the basic theory of PE brushes seems to be rather well understood by now, applications (several of which have been highlighted in the previous section) have started to emerge rather recently. In real life applications, isotropic and anisotropic architectures are in demand. Such architectures can be fabricated by grafting surfaces with PE brushes; the responsiveness of these PE brushes to environmental stimuli (such as changes in temperature, pH, ionic strength, *etc.*) will ensure potential applications of the PE-grafted surfaces in a large number of disciplines ranging from device technology to drug delivery systems. Mixed PE brushes have been identified as promising candidates for the fabrication of “smart” and adaptable surfaces.³⁵⁵ These surfaces, with triggerable phase transition behaviour, are relevant in the biological field for protein affinity separation and as switches in nano- and microfluidic devices. Specifically, devices with stimulus-responsive fusion proteins that can be reversibly addressed to the patterned surface through hydrophobic interactions are also being tested.³⁵⁶

The strong swelling behaviour of PE brushes, in the presence of suitable ions, could be used in the fabrication of nano-actuators. This is attributed to the fact that PE brushes can be switched from “soft matter” to “hard matter” in the presence of suitable ions.³⁵⁷ Another aspect that has recently emerged as a promising area of interest is the investigation of motion of nano-objects on polymer and PE brushes. PE brushes can be used to immobilize metallic nano-particles which in turn can be effectively employed as biocatalysts.³⁵⁸ Mechanical devices can be designed based on PE brushes since the switching-in-thickness of the brush, upon variation of an external parameter, is the essential ingredient for a valve. For example, if an AFM cantilever is coated on one side with a cationic brush, an asymmetric deflection response to an alternate voltage bias would be generated. This unique response of the AFM cantilever would be due to the presence of the asymmetric coating of the cationic brush on its outer surface. Similarly, an electrode coated by a polybase brush would provide tunable redox activities, due to the change in conformation of the brush that controls the transport of charged species to the electrode.

Potential applications are also being investigated for situations where there are interactions of PE brushes from multiple objects (*e.g.*, assembly of nanoparticles grafted with PE brushes). Such interactions can be vitally important to achieve desired

properties in a nanoparticle-augmented nanocomposite,³⁵⁹ for triggering of plasmonic hot spots by ensuring desired inter-particle nanoscale separation between PE-grafted nano cubes,³⁶⁰ for the formation of ultralong one dimensional nano structures,³⁶¹ for fabrication of functional meso-structures,³⁶² for devising novel mechanisms of adhesion of soft tissues,³⁶³ and many more.

9 Concluding remarks

We have reviewed the state of the art in theory, synthesis and applications of PE brushes. The review is divided into four different parts. In the first part, divided into 3 sections (Sections 2–4), we review the state of the art in the continuum modelling of polymer and PE brushes. We lay a special emphasis in identifying the situations where the electrostatic effects induced by the PE charges can be decoupled from the entropic and the excluded volume interactions of the PE brush molecules. The current understanding allows the appropriate modelling of the PE brushes with pH-dependent charge density only for such decoupled situations; the findings reveal highly interesting electrostatics of the PE brushes. These electrostatics, on the other hand, regulate the transport of ions and liquids at interfaces grafted with PE brushes. In the second part of the review, described by Section 5, we discuss some of the existing models to atomistically describe the polymer and PE brushes. In the third part of the review, described in Section 6, we review in detail the standard approaches for synthesizing polymer and PE brushes; we specially emphasize on how such grafting is done for nanoparticles and nanochannels. In the last part (covered in Sections 7 and 8), we pinpoint some of the key present and future applications of PE brushes. In a nutshell, in this review we have specifically emphasized on how distinct activities related to PE brushes have started and then evolved over time.

This review primarily addresses the most simplistic situation, where the polymer or PE brushes are homo-polymer brushes grafted on solid, planar interfaces and being present in a “good” solvent. Of course, we do provide some discussions for other cases (*e.g.*, polymer or PE brushes in “poor” solvent, non-solid, non-plane interfaces, *etc.*); however, we refrain from detailed analysis of these scenarios.

To summarize, this review is meant to provide an enlightened understanding of the behaviour of PE brushes and PE-brush-grafted interfaces. These brushes and interfaces are ubiquitous in a large number of disciplines ranging from biology and nanotechnology to health and energy. Their understanding has gained massive importance over the last few years; our review provides a brief window into some of these key developments. We particularly feel that this review will be extremely helpful for people who are beginners in the field.

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